Byung Hyun Lee

AI Researcher

Domain-Specialized Models & Selective Knowledge Updates

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KEY HIGHLIGHTS RESEARCH SPECTRUM

Analyzing **domain-specific images** and **learning continually without forgetting**.

INSPECTION

Find & classify defects with little labeled data

- **Industrial AI**: few-shot defect seg. (Samsung DS)
- **Agentic Defect Analysis**
- **Pathology**: tumor detection, slide labels only

METROLOGY-LIKE LOCALIZATION

Learning segmentation without pixel labels

- **HYPERION**: fine-tuned FM for tumor segmentation
- **CoMEL**: localization kept across new organs

MODEL UPDATE & EFFICIENCY

Efficiently learning new processes

- **Selective knowledge updates**: add or erase knowledge without forgetting
- **6 top-conference papers** as (co-)first author
- **Speculative Encoding**: 3× faster, 6× less memory

LLM POST-TRAINING & AGENTS

Make LLMs reliable and efficient

- **HyperCLOVA X**: ~2× faster inference, +20% throughput
- **MLLM model merging** (NeurIPS 2026 Spotlight)
- **LLM / VLA post-training** (ICLR 2027 sub.)

DOMAINS

Domain-Specialized AI | LLM · NAVER Cloud | Medical · Gigapixel Images | Robotics · Samsung Research, KETI | Agentic AI

KEY HIGHLIGHTS ACHIEVEMENTS & EXPERIENCE

8 **Top-tier AI papers**
NeurIPS · CVPR · ICCV · ICLR · AAAI

7 **First / co-first author**
among top-tier papers

4 **ICLR 2027 submissions**
3 as co-first author

7 **Industry & gov. projects**
3 as project leader

Research Recognition

- NeurIPS 2026 **Spotlight** · AAAI 2024 **Oral** · CVPR 2026 **Compute Transparency Champion Award**
- MICCAI 2025 REG **2nd place (team leader)** · CVPR 2026 **Outstanding Reviewer** · Qualcomm Fellowship **Finalist**

Projects

- **Project leader** — **SD Biosensor** · **MSIT · NIPA** (gigapixel histopathology whole slide image) · **KETI** (world VLA model)
- **Samsung DS** (defect segmentation) · **Samsung Research** (robot grasping) · **Ministry of Health and Welfare** (gigapixel whole slide image AI) · **US Air Force · IITP** (human-machine teaming)

Industry Experience

- **NAVER Cloud**, National AI Foundation Model Project — contributed to **HyperCLOVA X 8B Omni / 32B Think**

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03 Gigapixel Image Analysis Pathology slides as a high-resolution inspection problem

04 Selective Knowledge Updates Expanding & refining knowledge without forgetting

05 Erasing via Unlearning Removing target knowledge while keeping the rest

06 Additional Research & Future Direction Image restoration · agentic analysis · conclusion

PROFILE



Byung Hyun Lee

AI Researcher

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https://hyun1a.github.io

EDUCATION

Ph.D., Electrical & Computer Engineering

Seoul National University | 2021.03 – 2027.02 (expected)

B.S., Electrical Engineering

UNIST | 2015.03 – 2018.08 | GPA 4.04 / 4.30

EXPERIENCE

Visiting Student Researcher, NAVER Cloud

National AI Foundation Model Project | 2025.10 – 2026.02

RESEARCH INTERESTS

Models & Domains

- Domain-specialized Models
- Industrial Inspection
- Multimodal LLMs · VLA
- LLM Reasoning
- Diffusion Models

Selective Knowledge Updates

- Continual Learning
- Concept Erasure · Unlearning
- Training-free Weight Updates
- Post-training · RL

PUBLICATIONS

8
Top AI conference
7 as first / co-first

4
Under review
ICLR 2027 · 3 co-first

4
Workshop / short
CVPR-W · ECCV-W · ICASSP

2
Journal papers
IEEE Access · PLOS One

2
Technical reports
HyperCLOVA X 8B · 32B

1
Patent
Granted in US · KR

PUBLICATIONS FIRST / CO-FIRST AUTHOR

TOP AI CONFERENCE (SELECTED PAPERS)

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs **NeurIPS 2026 (Spotlight)***
- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models **CVPR 2026***
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis **ICCV 2025**
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation **CVPR 2025***
- Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate **ICLR 2025***
- Doubly Perturbed Task-Free Continual Learning **AAAI 2024 (Oral)**
- Online Continual Learning on Hierarchical Label Expansion **ICCV 2023***

* Equal contribution

UNDER REVIEW · PREPRINT

- HYPERION: Hyperbolic Grounding for Histopathological Slide Representations **ICLR 2027 submitted***
- Self-Distilled Reinforcement Learning for Language-Robust VLA Models **ICLR 2027 submitted***
- S3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation **ICLR 2027 submitted***
- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis **Preprint 2026***

JOURNAL

- Towards Accelerating Model Parallelism in Distributed Deep Learning Systems **PLOS One 2023***

SHORT PAPER · DOMESTIC

- Empirically Accelerating Scaled Gradient Projection Using DNN for Inverse Problems **ICASSP 2021**
- Selective Concept Erasing for Safe Diffusion Models (Best Poster Award) **KSPC 2024**

PUBLICATIONS CO-AUTHOR · TECHNICAL REPORTS · PATENTS

CONFERENCE · UNDER REVIEW

- All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters for Specific Degradations **CVPR 2023**
- RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity **ICLR 2027 (sub.)**
- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation **arXiv 2025**

JOURNAL

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation **IEEE Access 2025**

TECHNICAL REPORT · HYPERCLOVA X TEAM

- HyperCLOVA X 8B Omni / 32B Think **2026**

WORKSHOP · DOMESTIC

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models **CVPR Workshop 2026**
- Expert Classifier Ensemble-based Post-processing Correction for Unbiased Scene Graph Generation **IPIU 2024**
- Mobile AI & AIM 2022 Challenge Reports (quantized super-resolution; depth estimation) **ECCV Workshop 2022**
- Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction **KSPC 2022**

PATENT

- All-in-one image quality improvement model **US 2026 · KR 2025 (granted)**

PROJECTS & AWARDS

PROJECTS

KETI (Project Leader) — **2026 – present**
Uncertainty-aware World VLA Model

MSIT · NIPA (Project Leader) — **2026**
Hyperbolic Pathology FM for Gigapixel WSI

SD Biosensor (Project Leader) — **2025**
Industry–academia collaboration

US Air Force · IITP — **2025 – present**
AI for Human-Machine Teaming

Samsung DS — **2024 – 2025**
Industrial Defect Segmentation

Ministry of Health and Welfare — **2023 – 2024**
AI System for Gigapixel Histopathological WSI Analysis

Samsung Research — **2022 – 2023**
Robot Grasping

AWARDS & HONORS

CVPR Compute Transparency Champion Award — **2026**
Erasing Thousands of Concepts (ETC)

CVPR Outstanding Reviewer (Top 5%) — **2026**

Qualcomm Innovation Fellowship Korea — **2025**
Best Paper Finalist

MICCAI REG Challenge — 2nd Place (Team Leader) — **2025**
Report generation from gigapixel WSIs · 26 teams

Apple Scholars in AI/ML PhD Fellowship — **2025**
Institutional Nominee

Yulchon AI Star Scholarship — **2025**

KSPC Best Poster Presentation Award — **2024**

ACADEMIC SERVICE

Reviewer: ICLR · NeurIPS · CVPR · ECCV · AAAI · IEEE TIP · PR

RESEARCH OVERVIEW

Domain-Specialized Models & Selective Knowledge Updates

Industrial & Medical

Analyzing huge images

DOMAIN-SPECIALIZED

Few-shot Defect Seg.	Samsung DS 2025
CoMEL	ICCV 2025
HYPERION	ICLR 2027 sub.
Speculative Encoding	Preprint 2026
Tree-of-Experts	MICCAI REG 2025 · 2nd

Expanding & Refining

Add knowledge, no forgetting

CONTINUAL LEARNING, MODEL MERGING, REINFORCEMENT LEARNING

HLE	ICCV 2023
DPCL	AAAI 2024 Oral
DCAT	NeurIPS 2026 Spotlight
S^3-OPSD	ICLR 2027 sub.
MOSAR (VLA)	ICLR 2027 sub.
HyperCLOVA X	NAVER Cloud 2026

Erasing

Remove sensitive knowledge

CONCEPT ERASURE

CPE	ICLR 2025
GLoCE	CVPR 2025
ETC	CVPR 2026 · Award

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05 Erasing via Unlearning — Removing target knowledge while keeping the rest

06 Additional Research & Future Direction — Image restoration · agentic analysis · conclusion

Training-free Few-shot Industrial Defect Segmentation

Samsung DS | Industry project · Dec 2024 – Mar 2025 DOMAIN-SPECIALIZED AI

One labeled example is enough to segment a new defect type — no training.

PROBLEM

Anomaly detection **cannot name defect types**; trained models **need many masks**

IDEA

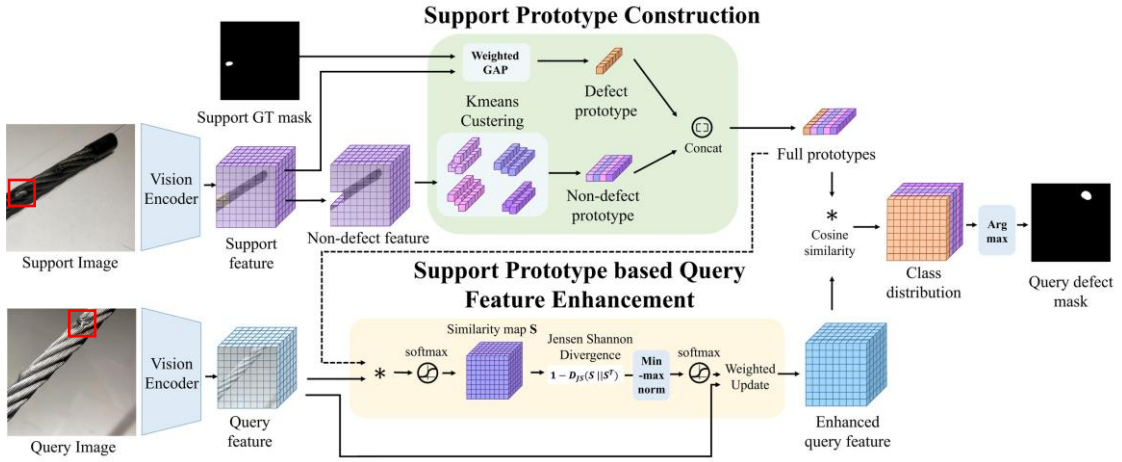
Match the example's defect features in new images with **defect + background prototypes**

RESULT

22.0 mIoU — 2× the training-free baseline, ≈ trained SOTA (22.5)

APPLICABLE TO INSPECTION & METROLOGY

Segment a **new defect type** from one reviewed example — **no retraining**



Defect / background prototypes → query enhancement

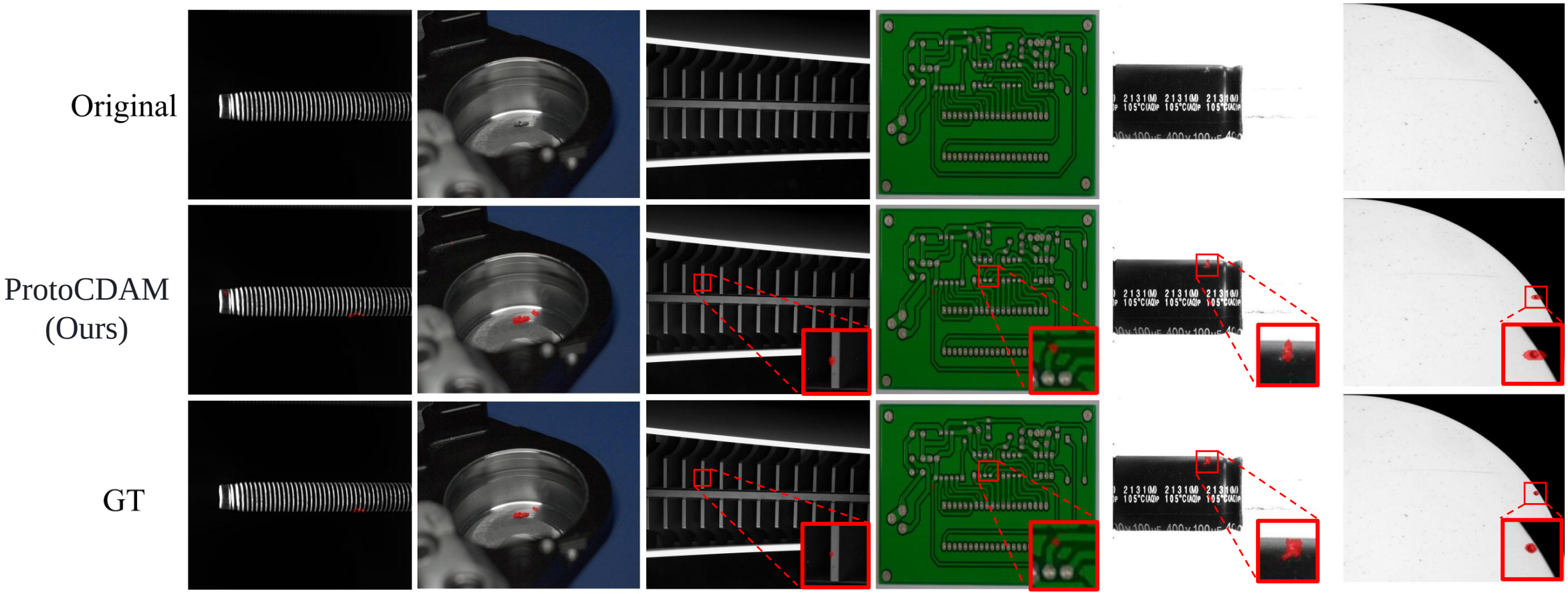
	Method	VISION V1	Defect Spectrum
Training	SOFS (SOTA)	22.5	39.1
Training-free	SegGPT	11.1	31.5
Training-free	**ProtoCDAM (Ours)**	**22.0**	**36.5**

1-shot defect segmentation, mIoU (%)

Training-free Few-shot Industrial Defect Segmentation

Samsung DS | Industry project · Dec 2024 – Mar 2025

DOMAIN-SPECIALIZED AI



1-shot results on unseen industrial objects (red = defect)

On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

Commercial agents plan and call tools;
only an on-premise agent sees the in-house data.

PROBLEM

Commercial LLMs excel with MCP tools, but **industrial data such as wafer maps and SEM images cannot leave the fab**

IDEA

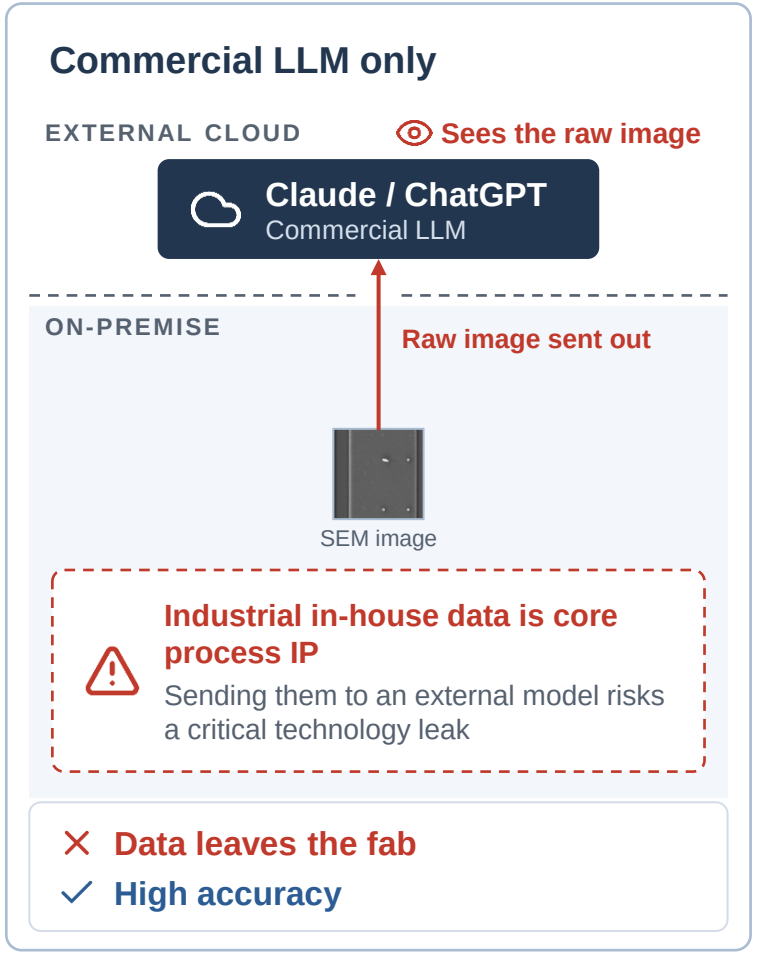
Run an **on-premise agent as an MCP tool** and the commercial LLM only calls it and **never sees the image**

CHALLENGE

Since defects are tiny, **open on-premise models often miss them** and fall short of commercial models

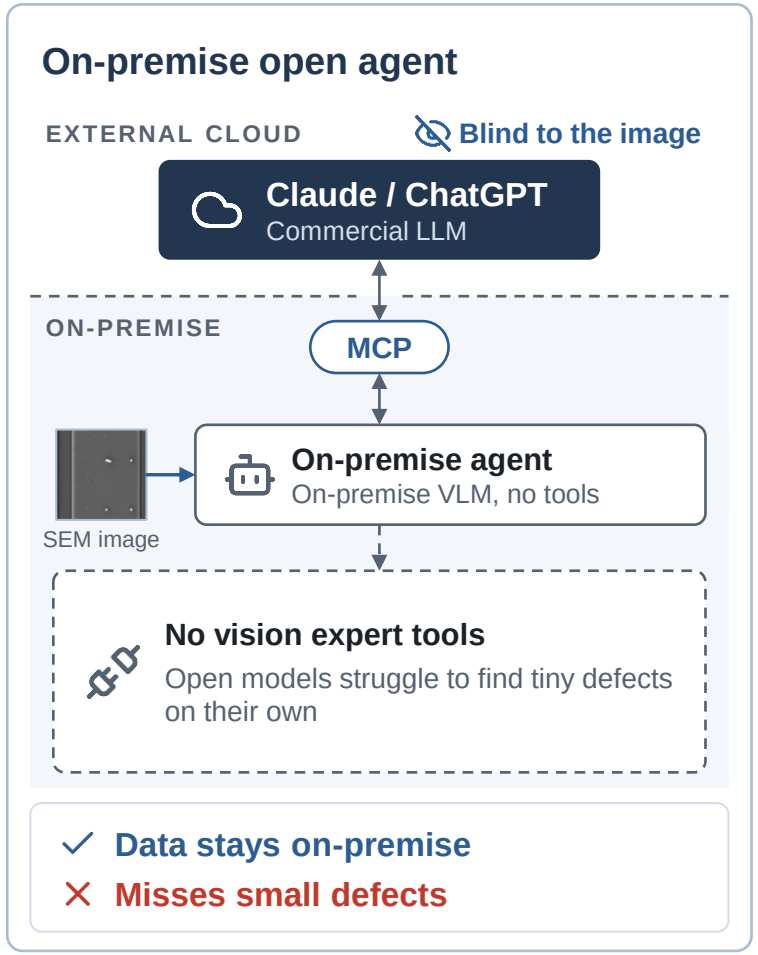
SOLUTION

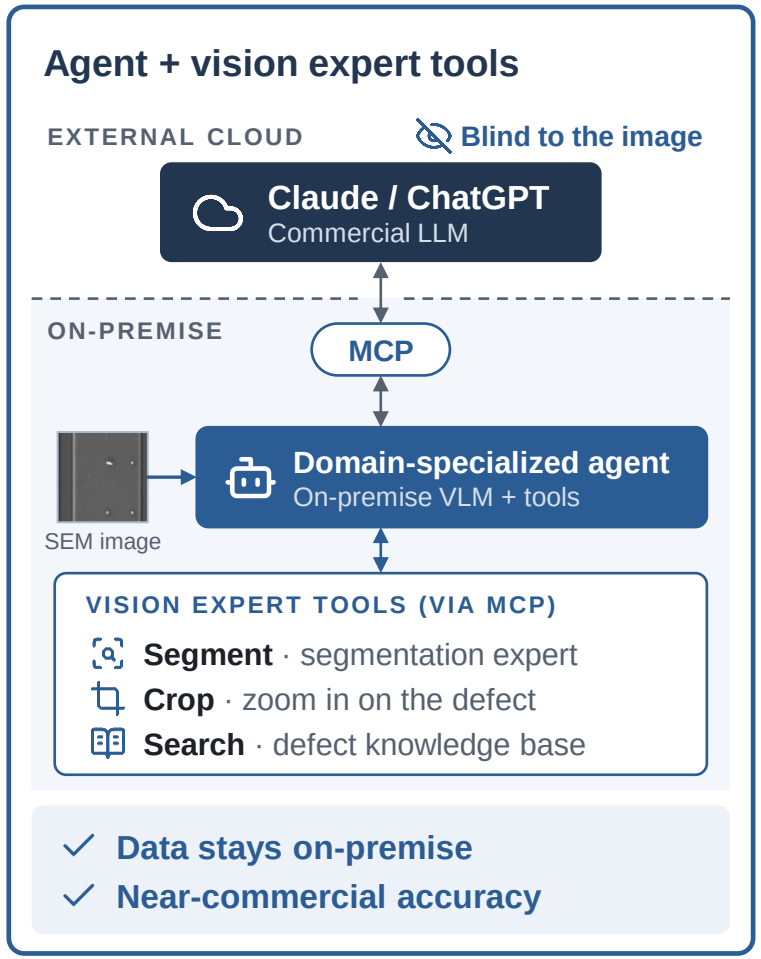
Give the agent **vision expert tools** (segmentation, crop, search) → **near-commercial performance (DefectAgent)**



On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI





DefectAgent

On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

Two tasks on industrial inspection images; **every answer must also show where the defect is.**

Defect VQA

Short visual question answering

TASK Answer one short question and **box the defects** the answer refers to

OUTPUT answer + defect region

Grounded Inspection Report

Defect listing + inspection report

TASK List every defect with localization, then **write a report and decide PASS / FAIL**

OUTPUT report text + defect region

DATA

SEM wafers · IC SEM · industrial parts

Carinthia-S, MIIC, VISION, MVTec AD

OURS

DefectAgent:
Qwen3.6-35B (public) + ProtoCDAM

COMPARED

Claude Sonnet / Opus 5.5 / Qwen3.6-35B (public)

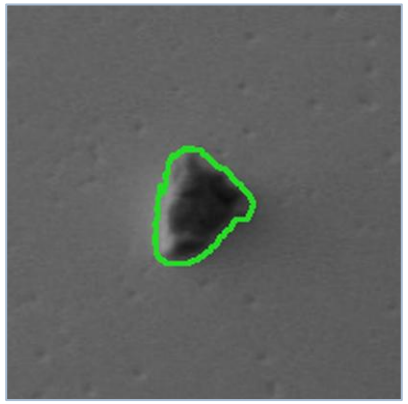
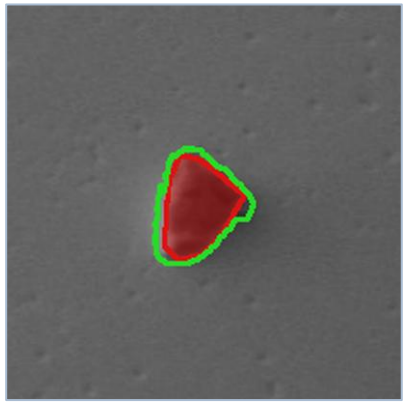
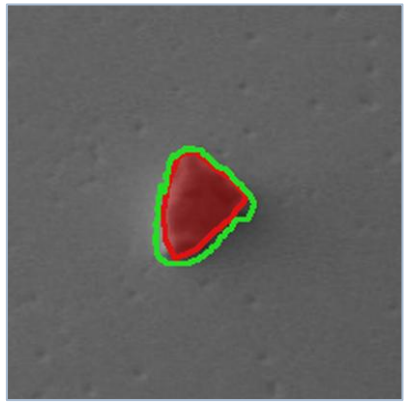
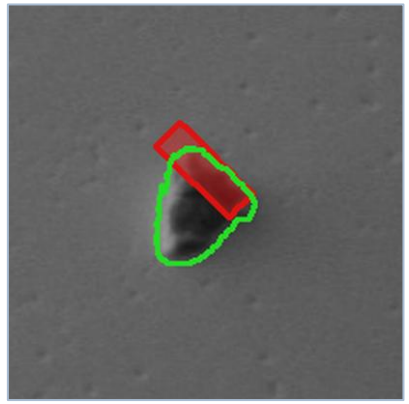
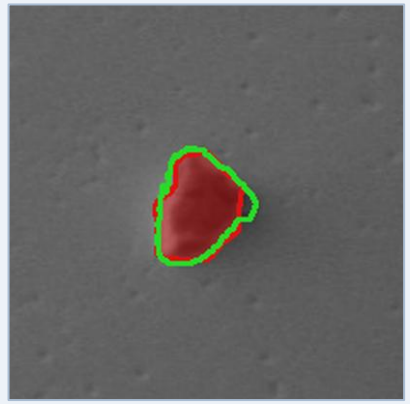
On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

DEFECT VQA **Carinthia-S · SEM wafer**

Q Is there any defect in the image? Answer “yes” or “no”.

ground truth model prediction

Ground truth	Claude Sonnet 5.5	Claude Opus 5.5	Qwen3.6-35B	DefectAgent (Ours)
				
Answer: yes	✓ **yes** IoU 73.5	✓ **yes** IoU 72.9	✓ **yes** IoU 29.5	✓ **yes** **IoU 82.7**

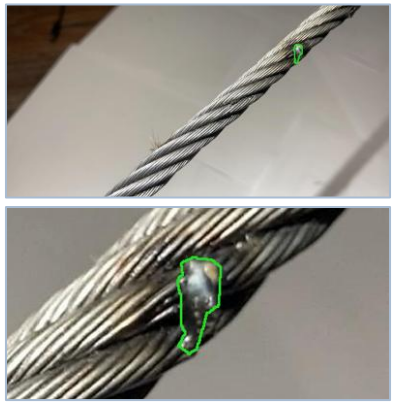
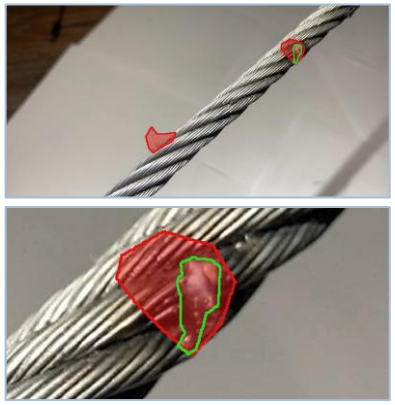
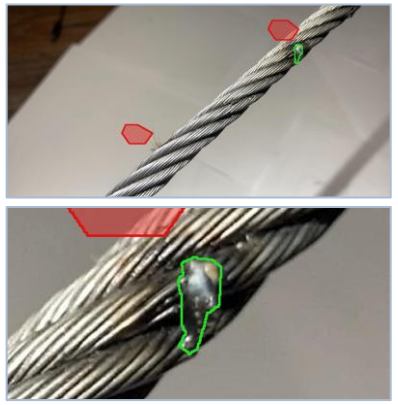
On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

DEFECT VQA **VISION · steel cable**

Q Does the part PASS or FAIL?

ground truth model prediction

Ground truth	Claude Sonnet 5.5	Claude Opus 5.5	Qwen3.6-35B	DefectAgent (Ours)
				
Answer: PASS	✕ **FAIL** IoU 18.8	✕ **FAIL** IoU 14.7	✓ **PASS** IoU 0.0 **Hallucination**	✓ **PASS** **IoU 71.4**

On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

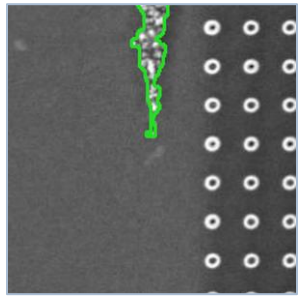
DefectAgent | 2026 DOMAIN-SPECIALIZED AI

GROUNDED INSPECTION REPORT **MIIC · IC SEM**

Task List every defect with its location, write a report and decide PASS / FAIL

ground truth model prediction

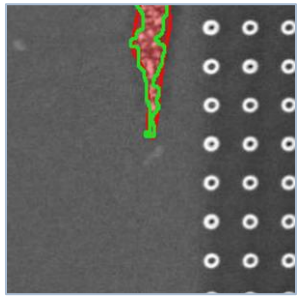
Ground truth



Decision: FAIL

*One defect region at the **top-middle**, touching the top edge. Area **0.709 %** of the image (major), above the 0.1 % limit.*

Claude Sonnet 5.5

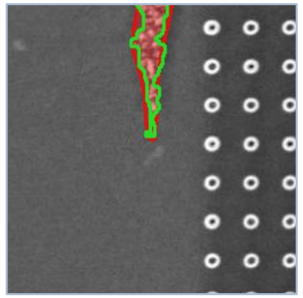


✓ **FAIL**

IoU 73.3

*One defect [D1] is present in the image. It is a vertical streak of bright granular debris at the top-center, left of the contact-hole array. It touches the top image edge. Its shape is irregular and line-like, and it covers about 1100 px, or roughly **0.42 %** of the image, which is major. This is above the 0.1 % limit. **Decision: FAIL***

Claude Opus 5.5

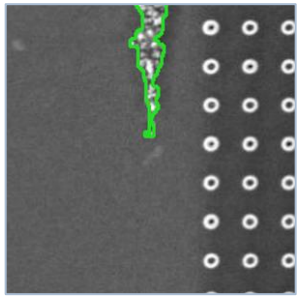


✓ **FAIL**

IoU 70.6

*There is one defect of type [D1], a cluster of bright particles in the top-left area, to the left of the contact array. It touches the top edge of the image and runs downward as a stretched, irregular streak to about y=122 px. It is about 38 x 122 px across, with an estimated particle area of about 1300 px, or about **0.5 %** of the image. That makes it a major defect. The total defect area is above the 0.1 % limit, so the part fails. **Decision: FAIL***

Qwen3.6-35B

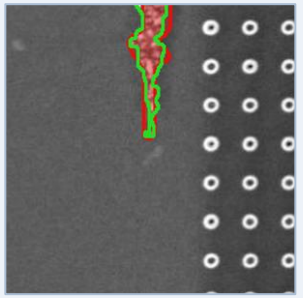


✗ **PASS**

IoU 0.0

*The inspection of the provided SEM image **reveals a clean integrated-circuit layout**. The array of circular vias is uniform and intact, with no visible breaks, shorts, or foreign material. **There are no defects detected in the image. Decision: PASS***

DefectAgent (Ours)



✓ **FAIL**

IoU 71.0

*The inspection identified one defect region, [D1], located in the **top-middle** of the integrated-circuit layout. This defect is line-like in shape and touches the top and left edges of the image. It covers a major area of approximately **0.86% (2267 pixels)** of the total image area. Since the total defect area exceeds the 0.1% threshold, the part is considered to have failed the inspection. **Decision: FAIL***

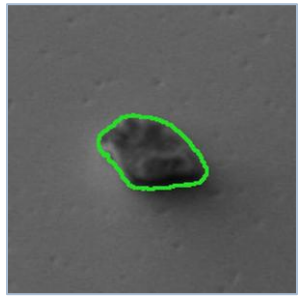
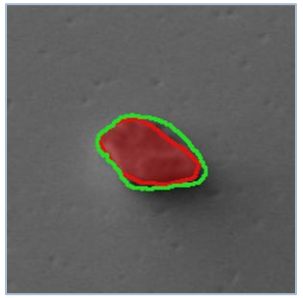
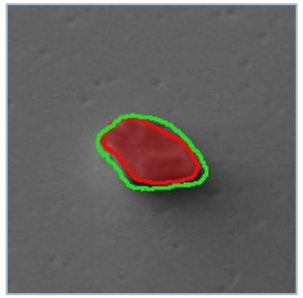
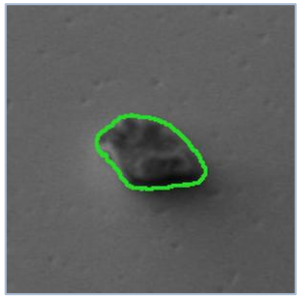
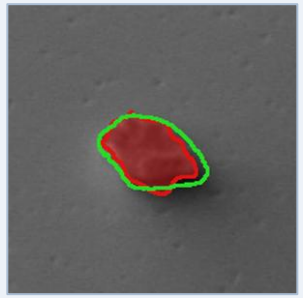
On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

GROUNDED INSPECTION REPORT **Carinthia-S · SEM wafer**

Task List every defect with its location, write a report and decide PASS / FAIL

ground truth model prediction

Ground truth	Claude Sonnet 5.5	Claude Opus 5.5	Qwen3.6-35B	DefectAgent (Ours)
				
Decision: FAIL	✓ **FAIL**	✓ **FAIL**	✕ **PASS**	✓ **FAIL**
	IoU 76.8	IoU 73.2	IoU 0.0	**IoU 77.3**
One defect region at the **center**. Area **1.679 %** of the image (major), above the 0.1 % limit.	One defect was found, a single irregular particle-like feature [D1] near the center of the image. It does not touch the image edge. It measures about 86 x 56 px and covers about 2900 px, roughly 1.3 % of the image area, which is major. Its shape is irregular, a blob with a rough outline. The total defect area is well above the 0.1 % limit. **Decision: FAIL**	There is one defect of type [D1] (defect) in Image 1. It is a raised, particle-like piece of debris with an irregular outline, located in the center of the image at about x 227-310, y 153-210. It does not touch the image edge. Its estimated area is about 2,600 px, roughly 1.1 % of the image area, so it counts as major. That is above the 0.1 % failure threshold, so the part fails. **Decision: FAIL**	**No defect is visible.** The image shows a semiconductor wafer surface with a small, irregular particle or debris located in the center. This feature **does not meet the criteria for a manufacturing defect** as defined by the inspection rules. **Decision: PASS**	The inspection of the semiconductor wafer surface identified one defect, [D1]. This defect is a major irregular blob/spot located at the **center of the image**, with an area of **1.3641% (3143 px)**. It does not touch the image edge. Since the total area of defects (1.3641%) exceeds the 0.1% threshold, the part fails the inspection criteria. **Decision: FAIL**

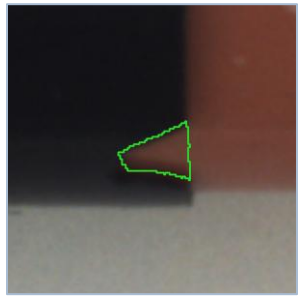
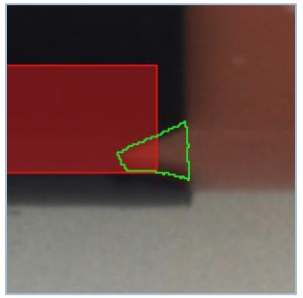
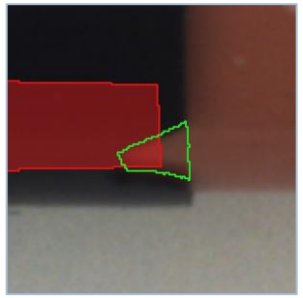
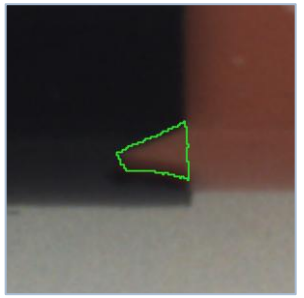
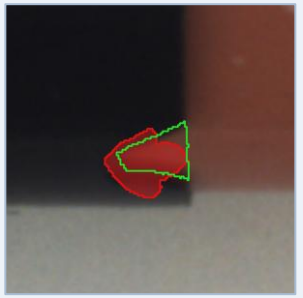
On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

GROUNDED INSPECTION REPORT **MVTec AD · capsule**

Task List every defect with its location, write a report and decide PASS / FAIL (FAIL if defect area ≥ 0.1 %)

ground truth model prediction

Ground truth	Claude Sonnet 5.5	Claude Opus 5.5	Qwen3.6-35B	DefectAgent (Ours)
				
Decision: PASS	✕ **FAIL**	✕ **FAIL**	✓ **PASS**	✓ **PASS**
	IoU 1.9	IoU 2.2	IoU 0.0	**IoU 44.0**
*One **crack** at the center of the capsule. Area **0.068 %** of the image (minor), below the 0.1 % limit.*	*One defect type is present, **a faulty imprint [D1]**, and there is one instance. It is in the bottom-center of the image, on the black capsule half, where the logo imprint is missing and only faint smudges remain. It covers about 12,400 px, **roughly 1.2 % of the image**, which is major. Its shape is an irregular rectangular patch, and it does not touch the image edge. This is well above the 0.1 % limit. **Decision: FAIL***	*There is one defect on this capsule, [D1], **a faulty imprint**. The 'actavis' logo is missing from the lower part of the dark cap, in the middle-left of the image just left of the joint between the two halves. The affected area is an irregular, wide band about 245 x 60 px, or **roughly 1.1 % of the image area**. It does not touch the image edge. This is above the 0.1 % limit, so the part fails. **Decision: FAIL***	***No defect is visible.** The capsule appears intact with no cracks, scratches, or other anomalies detected. **Decision: PASS***	*The inspection identified one defect on the capsule, classified as **a crack [D1]**. This defect is located at the center of the capsule, specifically at **the seam between the black and orange halves**. It appears as a small, irregular blob/spot and does not touch the image edges. The defect is minor, covering an area of **0.0937% of the image (937 pixels)**. Since the total area of all defects is less than the 0.1% threshold required for failure, the part is considered acceptable. **Decision: PASS***

On-Premise Agents with Vision Expert Tools for Secure Defect Analysis

DefectAgent | 2026 DOMAIN-SPECIALIZED AI

With ProtoCDAM, the on-premise model is **best or tied-best on 6 of 9 metrics** and close to Claude on the rest.

	GROUNDING (BOTH TASKS)			GROUNDED INSPECTION REPORT					DEFECT VQA
Model	**Box mIoU**	**Box F1@0.5**	**Mask mIoU**	**Fact F1**	**Halluc. % ↓**	**Report F1**	**Report score**	**Decision acc.**	**Accuracy**
Claude Sonnet 5.5	43.6	56.6	37.2	**53.5**	8.0	23.0	77.9	**87.5**	80.0
Claude Opus 5.5	46.1	56.8	38.9	48.5	18.5	**27.4**	76.3	79.2	**84.2**
Qwen3.6-35B-A3B	18.9	11.1	13.2	7.7	63.6	14.4	27.7	12.5	68.9
DefectAgent (Ours)	**50.4**	**62.3**	**40.6**	52.5	**3.8**	25.9	**78.5**	**87.5**	81.4

Bold = best in column · ↓ lower is better · Grounding is class-agnostic, on items with at least one defect

INDUSTRY EXPERIENCE

NAVER Cloud · HyperCLOVA X

Visiting Student Researcher

NAVER Cloud | Oct 2025 – Feb 2026

Korea's National AI Foundation Model Project

~2×
faster inference, kept stable

+20%
training tokens per unit time

Pre-training
Large-scale data → base model

HyperCLOVA X 8B Omni
HyperCLOVA X 32B Think

Post-training
SFT · RL → instruction & reasoning

CONTRIBUTION

- ✓ Training & inference **stability**
- ✓ **+20% throughput** (batching)

Inference & deployment
Serving the model online

CONTRIBUTION

- ✓ Stable at **~2× faster inference**
- ✓ Code for **online deployment**

APPLICABLE TO LLM / AGENTS

Training and serving **in-house LLMs and engineering assistants** — stable, fast and efficient

PROJECTS INDUSTRY · GOVERNMENT · CHALLENGE

AS PROJECT LEADER / TEAM LEADER

SD Biosensor
Project Leader · 2025

Industry–academia collaboration

Industrial

MICCAI 2025 REG
Team Leader · 2nd / 25

Report generation from gigapixel WSIs

Medical

MSIT · NIPA
Project Leader · 2026

Pathology FM for gigapixel WSI

Medical

KETI
Project Leader · 2026 ~

Uncertainty-aware world VLA model

Physical · Agentic

AS PARTICIPATING RESEARCHER

Samsung DS
2024 – 2025

Industrial defect segmentation

Industrial · Semiconductor

Samsung Research
2022 – 2023

Robot grasping

Physical

Korea MOHW
2023 – 2024

Gigapixel pathology AI system

Medical

US Air Force · IITP
2025 – present

AI for human-machine teaming

Agentic

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02 Industry Projects — Industrial Defect Analysis (Samsung DS) · LLM post-training (NAVER Cloud)

03 Gigapixel Image Analysis — Pathology slides as a high-resolution inspection problem

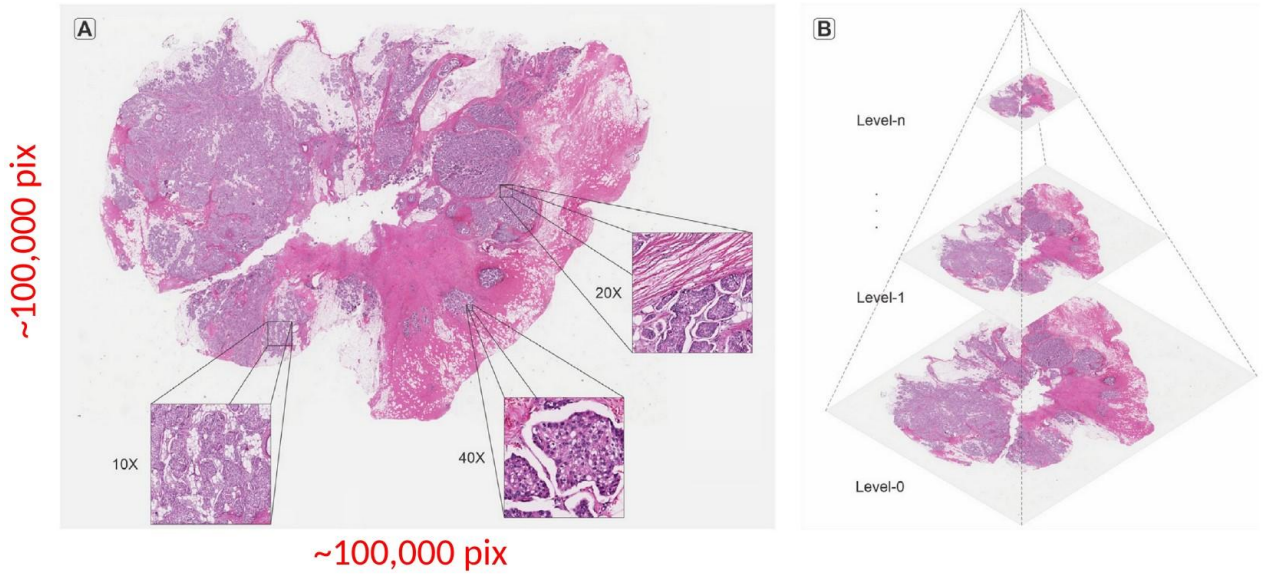
04 Selective Knowledge Updates — Expanding & refining knowledge without forgetting

05 Erasing via Unlearning — Removing target knowledge while keeping the rest

06 Additional Research & Future Direction — Image restoration · agentic analysis · conclusion

WHOLE SLIDE IMAGES WHAT THE DATA LOOKS LIKE

A tissue slide scanned at cell-level resolution — **one image ≈ 100,000 × 100,000 pixels**.



A whole slide image (left) and its zoom pyramid (right)

~10 B pixels in one slide (several GB)

10K–100K patches the slide is cut into

SEMICONDUCTOR ANALOGY

- **Slide ≈ wafer / SEM image**
- **Patch ≈ field of view**
- **Slide label ≈ wafer pass / fail**
- **Tumor ≈ defect**

WHOLE SLIDE IMAGES

WHAT AI MUST DO — AND WHY IT IS HARD

WHAT AI MUST DECIDE

Tumor detection / subtype classification

≈ Inspection

Tumor segmentation

≈ Metrology-like localization

Diagnosis

≈ Review report

CHALLENGE

Too large for a GPU

cut into tens of thousands of patches

Only a slide-level label

pixel-level masks are expensive and scarce

Data is split across institutions

patient data cannot leave each hospital

Lesions are tiny

often a small fraction of the slide

AI PIPELINE FOR A GIGAPIXEL IMAGE FOUR STEPS



1 Encode patches

Cut the slide into small tiles; a pretrained encoder describes each tile

→



2 Aggregate

Combine all tile features into one slide-level decision

→



3 Locate

Find the tiles behind the decision → a tumor map

→



4 Report

Turn the findings into a structured report

≈ Scanning

≈ Inspection (pass / fail)

≈ Metrology-like localization

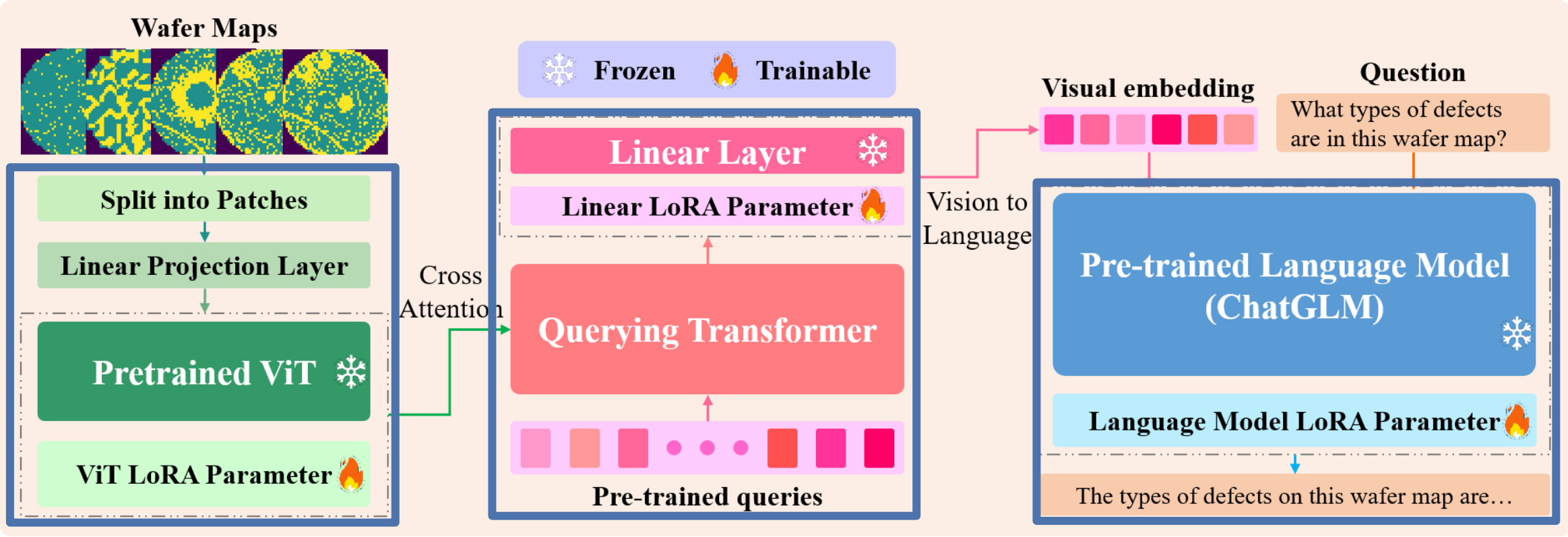
≈ Review report

AI PIPELINE FOR A GIGAPIXEL IMAGE

SAME STRUCTURE IN WAFER AI

A recent wafer-map MLLM uses similar pipeline: **encode patches** → **aggregate** → **report** (no locate step yet)

WamGLM: wafer-map defect MLLM (Gu et al., Applied Soft Computing)



CONTRIBUTION TO EACH STEP

STEP	BOTTLENECK	PROPOSED METHOD	RESULT
Encode patches	**~99% of inference time**	**Speculative Encoding** draft cheaply, verify selectively	**3× faster**, 6× less memory
Aggregate & Locate	**Only slide labels** **No pixel labels**	**CoMEL** new organs, no stored data	**+23.4%** localization IoU
		HYPERION no pixel labels needed	Dice **0.143 → 0.463**
Report	**Strict format**, < 10K slides	**Tree-of-Experts** one small expert per branch	**2nd place**, MICCAI 2025 REG

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

DOMAIN-SPECIALIZED AI & EXPANDING

Learns new organs one by one without storing patient data — and keeps tumor localization.

✕ **PROBLEM**

Storing past slides is **costly & risky**; prior methods **silently forget localization**

IDEA

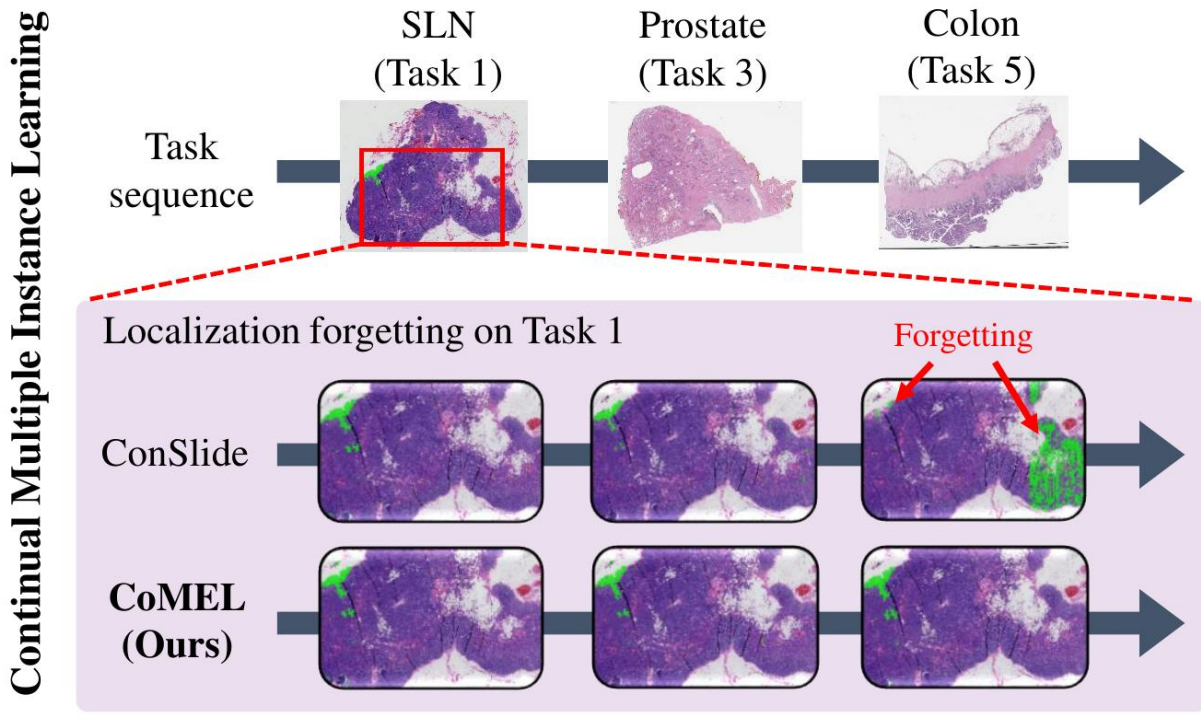
Add new knowledge in directions that **do not overwrite the old** (orthogonal LoRA)

✓ **RESULT**

Up to **+11.0% slide accuracy** and **+23.4% localization IoU**

APPLICABLE TO INSPECTION & METROLOGY

Learn a **new process or defect type** without erasing existing knowledge

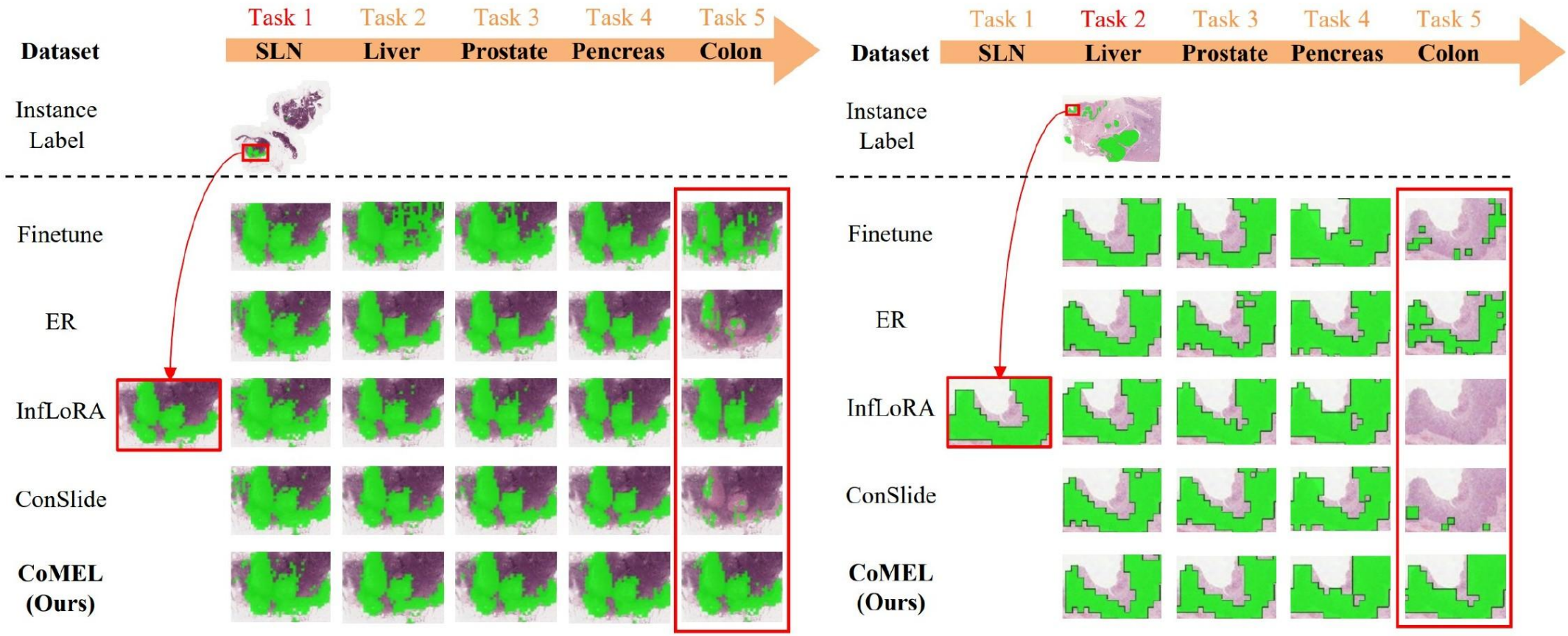


ConSlide forgets Task-1 tumors; CoMEL keeps them

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

DOMAIN-SPECIALIZED AI & EXPANDING



Tumor localization over the task sequence

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI

Training a model to point at the evidence for each pathology concept.

✕ **PROBLEM**

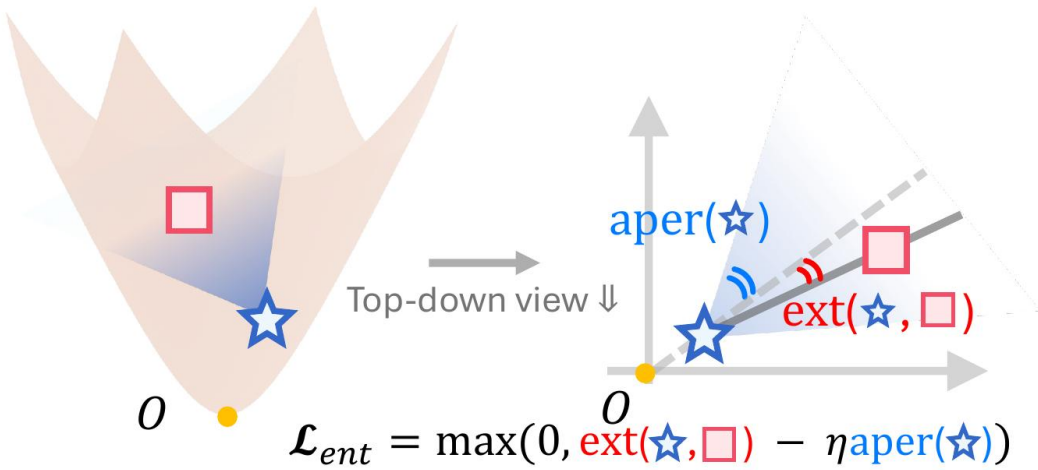
Slide encoders match slide ↔ report but **cannot show which region supports each concept**

IDEA

Fine-tune the slide encoder so slide & text embeddings form a **tree** (hyperbolic space)

✓ **RESULT**

Zero-shot tumor segmentation: avg. Dice **0.143 → 0.463**



Hyperbolic space ≈ a tree: general concepts at the center, specific evidence at the edge

APPLICABLE TO METROLOGY-LIKE LOCALIZATION

Outline regions that match a **text query** — **without pixel labels**

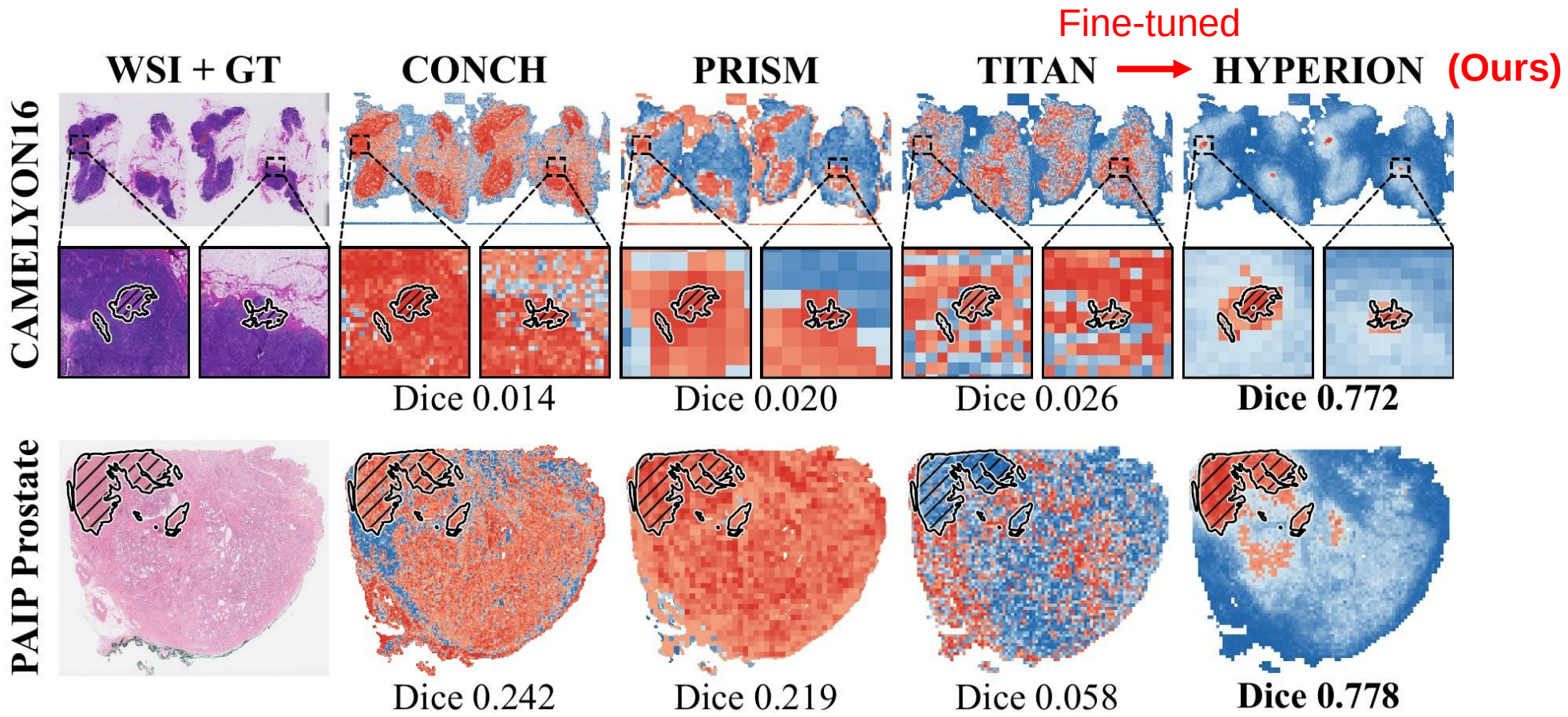
✕ **Flat (Euclidean)** embeddings squeeze the tree → **weak grounding**

✓ **HYPERION** keeps the tree → each sentence finds its region

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI

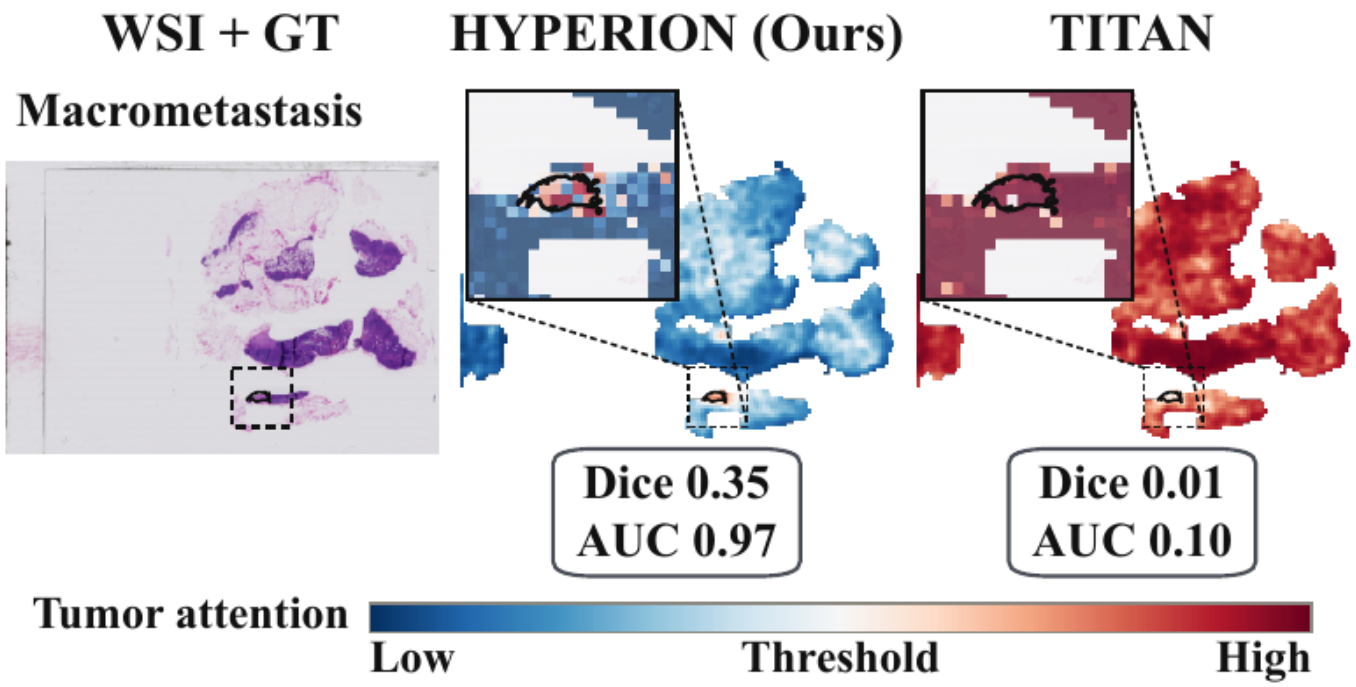


Zero-shot tumor segmentation vs. pathology foundation models

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI



MLLM answer (VQA)

Q: Which observed features are most significant for making a diagnosis?

✓ **HYPERION:** …disruption of lymph node architecture by these metastatic cells…

✗ **TITAN:** …atypical melanocytic cells… infiltration into the stroma…

MLLM + HYPERION finds the metastasis and answers correctly; the baseline fails

Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis

Preprint 2026 | Co-first author

DOMAIN-SPECIALIZED AI

Big-model patch encoding is the bottleneck — let a small model decide where to look.

PROBLEM

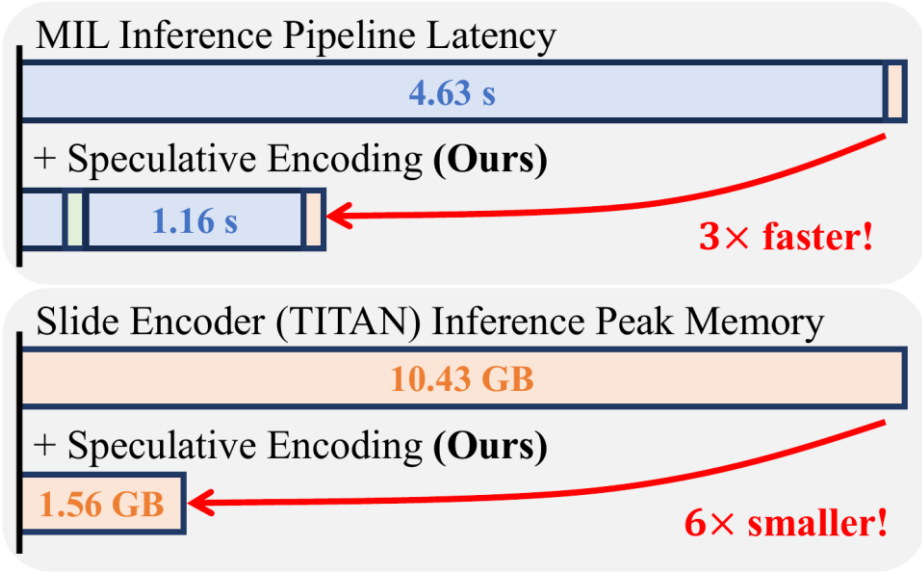
Patch encoding takes **~99% of inference time**; aggregation sets **peak memory**

IDEA

Draft all patches with a **small encoder**, re-encode only the **informative** ones

RESULT

3× faster · 6× less memory · >99% accuracy kept



Latency & peak memory: baseline vs. ours

APPLICABLE TO HIGH-THROUGHPUT

Scan everything cheaply, **analyze candidate regions in depth**

3×	6×	>99%
faster inference	**less peak memory**	**accuracy kept**
4.63 s → 1.16 s	10.4 → 1.6 GB	across 12 models

Tree-of-Experts for Report Generation in Histopathology

MICCAI 2025 · REG Challenge | 2nd place of the challenge · Team leader

DOMAIN-SPECIALIZED AI

Writes structured pathology reports with 64 small experts — 2nd place as team leader.

PROBLEM

Strict report format across 8 organs, **fewer than 10K training slides**

IDEA

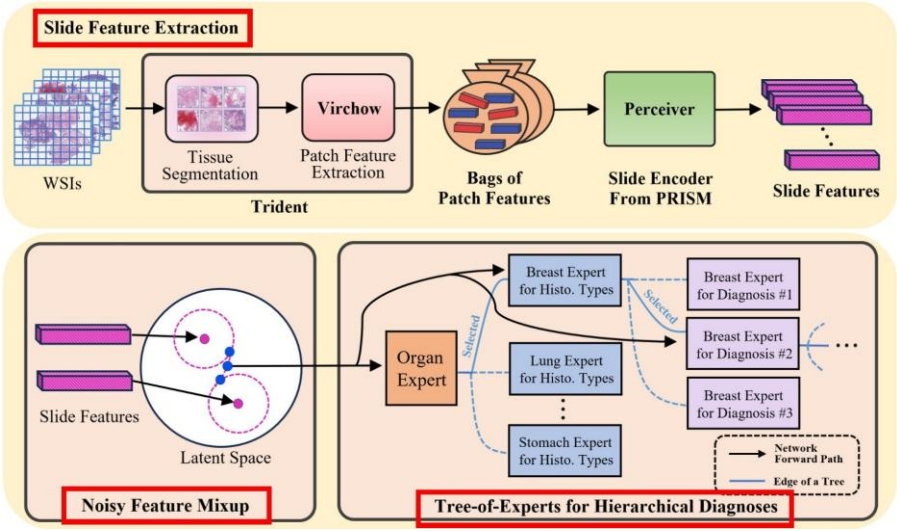
Predict the report as a **tree of choices** (organ → type → diagnosis), one expert per branch

RESULT

Ranking score **0.811** · **2nd place**, MICCAI 2025 REG

APPLICABLE TO REPORTING

Structured inspection / review reports from images, with little data



Frozen slide features → Tree-of-Experts

	Breast	Lung	Colon	Rectum	Stomach	Cervix	Prostate	Bladder	Avg.
Ranking	0.775	0.842	0.840	0.710	0.871	0.774	0.801	0.798	**0.811**
ROUGE	0.793	0.884	0.845	0.762	0.855	0.838	0.806	0.875	**0.832**

Validation scores per organ

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WHY KNOWLEDGE UPDATES

Foundation models now power products on three fronts — and each must keep updating its knowledge after deployment.

Agentic / Physical AI

LLM agents · VLA-driven robots

Online knowledge addition

Adapt quickly to new tasks, tools, environments and instructions

Vertical AI

Domain-specialized models

Domain knowledge addition

New organs, types, processes and defect types

Visual Generative AI

Text-to-image / video generation

Harmful knowledge deletion

Remove copyrighted, private or unsafe concepts on demand

INDUSTRY

Deployed models must adapt — without full retraining

Online knowledge addition agents & robots

Domain knowledge addition medical & industrial AI

RESEARCH

- **Catastrophic forgetting** in foundation models
- **Stability–plasticity** trade-off
- **Precise, scalable unlearning**

THE CHALLENGE: Update knowledge without forgetting

1 Full retraining

Retrain on all the data



Highly inefficient and costly

2 Naive knowledge update

Fine-tune only on the new data



Forgets existing knowledge (stability–plasticity dilemma)

3 Selective knowledge update

Update only the target, keep the rest



existing knowledge kept

Efficient and minimal forgetting

Intuition Teaching an inspector a new defect type: no retraining from scratch (1), no forgetting old types (2) — just add what is new (3).

Online Continual Learning (CL) on Hierarchical Label Expansion

ICCV 2023 | Co-first author

EXPANDING

Learns finer classes as labels get refined over time ("bird" → "sparrow").

PROBLEM

Labels grow coarse → fine, but standard continual learning **ignores the hierarchy**

IDEA

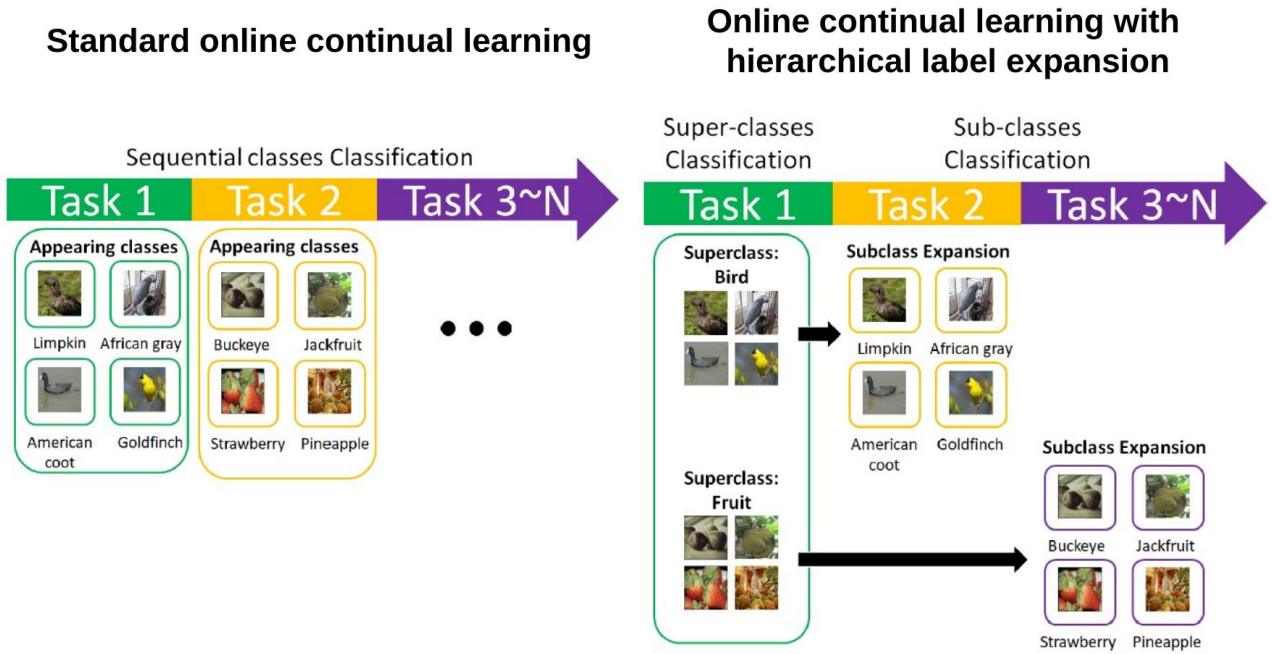
Use the **hierarchical memory**: pseudo-labels per branch + memory favoring new classes

RESULT

Best any-time accuracy on 3 benchmarks

APPLICABLE TO MODEL UPDATE

A defect class later **split into sub-types** — no relearning from scratch



Standard online CL (left) vs. hierarchical label expansion (right)

Online Continual Learning on Hierarchical Label Expansion

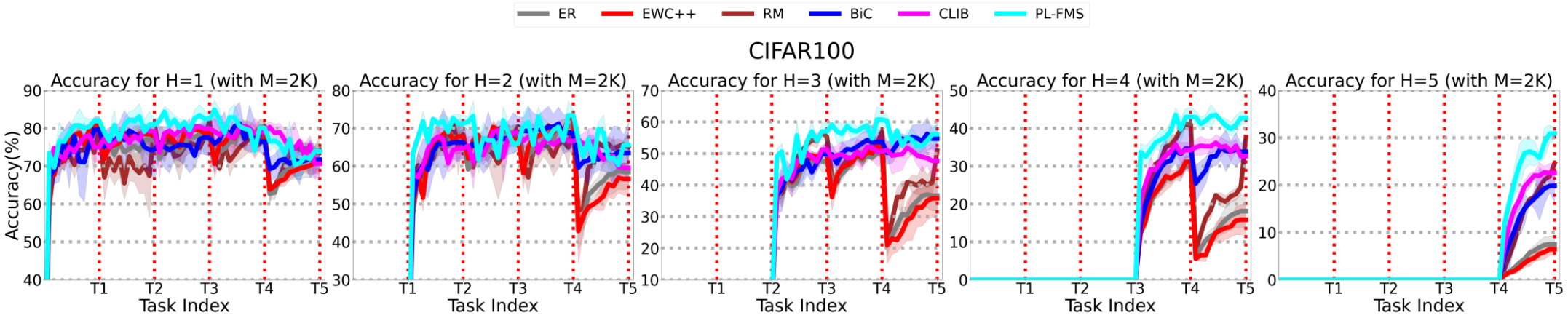
ICCV 2023 | Co-first author

EXPANDING

H#: hierarchical level

Methods	CIFAR100					iNaturalist-19						
	H=1	H=2	H=3	H=4	H=5	H=1	H=2	H=3	H=4	H=5	H=6	H=7
ER	71.5±4.44	58.4±4.58	36.6±4.78	18.1±4.28	7.47±1.61	84.9±6.03	84.9±0.68	59.8±15.5	29.3±3.28	17.8±3.95	13.0±3.95	1.50±0.77
EWC++	70.9±2.83	56.6±4.26	35.8±5.93	15.8±3.94	6.43±1.28	**87.4±2.38**	80.7±1.19	66.1±9.80	29.4±4.48	18.1±6.53	15.1±5.73	1.88±1.15
BiC	71.6±1.01	63.5±2.48	54.7±0.61	33.8±0.41	19.8±0.78	79.5±14.4	76.3±12.1	54.0±27.4	22.9±10.3	14.8±9.88	11.2±7.78	1.34±1.41
RM	74.2±3.99	65.0±4.18	50.9±1.40	37.6±0.60	24.5±2.54	74.0±5.57	69.7±4.21	54.4±2.20	40.7±1.15	37.4±0.85	35.1±0.44	11.3±0.33
CLIB	70.6±4.05	59.5±1.22	47.6±5.06	32.6±1.76	22.5±2.08	87.2±2.26	81.3±4.78	62.4±4.10	41.5±0.97	35.3±0.70	33.2±1.19	8.07±0.94
PL-FMS	**74.5±4.63**	**65.6±3.34**	**56.0±3.66**	**42.7±1.79**	**30.8±1.54**	86.1±3.15	**88.4±3.79**	**70.6±3.17**	**49.6±2.42**	**43.9±1.86**	**41.3±2.57**	**13.6±0.28**

Accuracy at each hierarchy level (H1 = coarsest)



Any-time accuracy per level (left: coarsest → right: finest)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING

Modeling the uncertainty of unseen future data → less forgetting, faster adaptation.

PROBLEM

Streaming data, no task boundaries: **today's update can hurt future learning**

IDEA

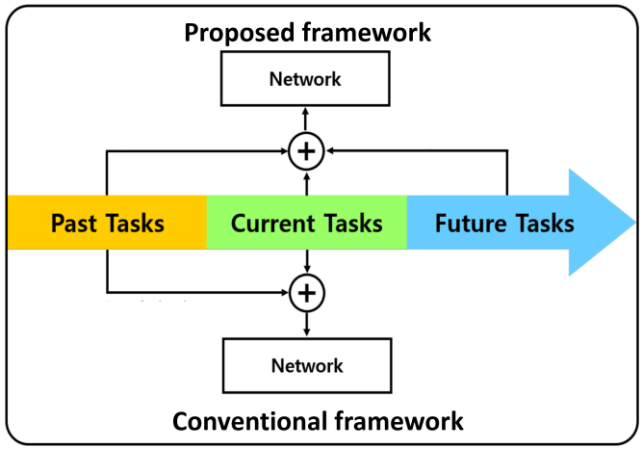
Train with random **perturbations of inputs and weights** (flatter minima)

RESULT

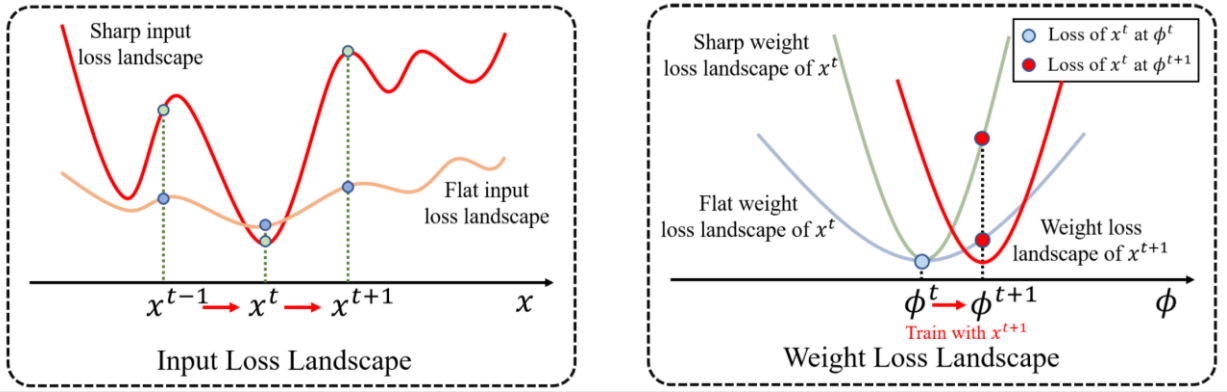
Best any-time accuracy on 3 benchmarks · **AAAI Oral**

APPLICABLE TO MODEL UPDATE

Production data that **drift continuously** (tools, recipes, lots)



Conventional vs. ours: also prepare for future tasks

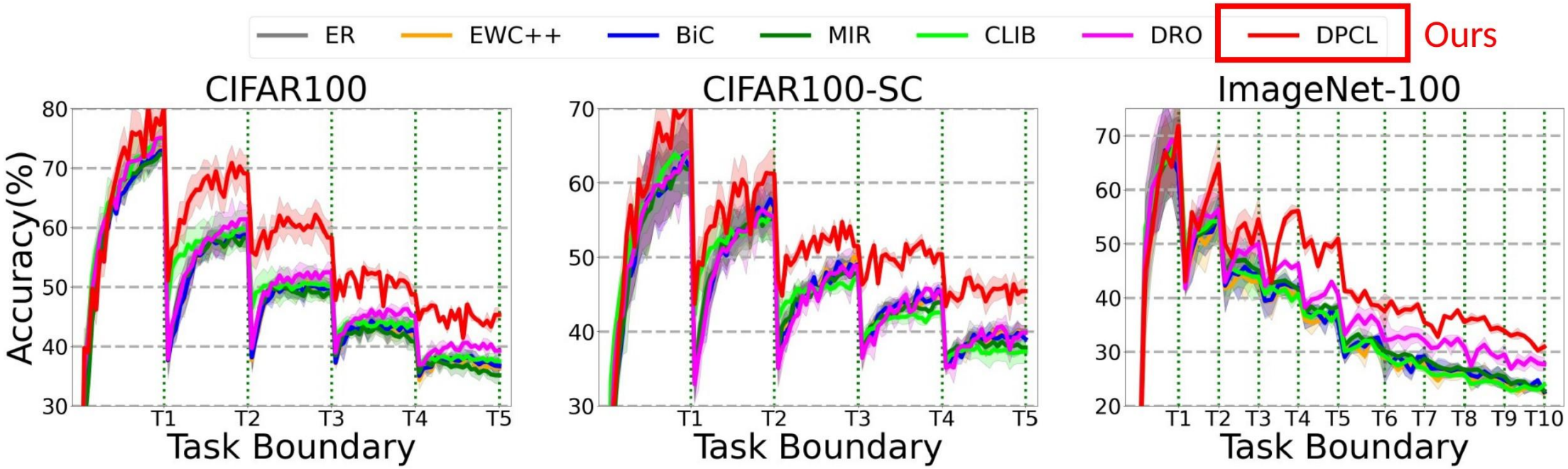


Sharp vs. flat loss landscapes (input / weight)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING



Any-time accuracy in class-incremental learning

Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author

REFINING

A strong vision MLLM helps weak audio / video MLLMs — no fine-tuning.

PROBLEM

Improving a weak modality needs **large data & compute**

IDEA

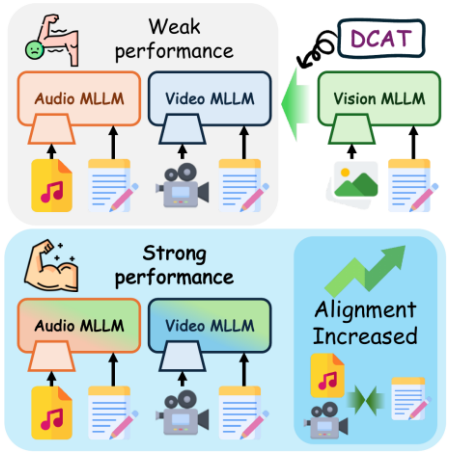
Transfer the strong model's text alignment with a **closed-form weight update**

RESULT

+29.55% audio · **+24.92% video** (average gain)

APPLICABLE TO LLM / MLLM

Boost an in-house MLLM on a **data-scarce image type**

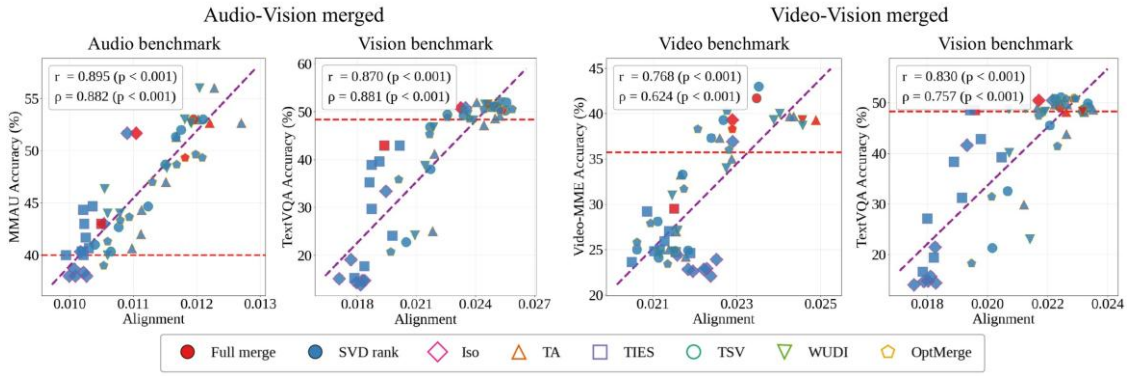


DCAT concept

+29.55% Avg. gain, audio benchmarks

+24.92% Avg. gain, video benchmarks

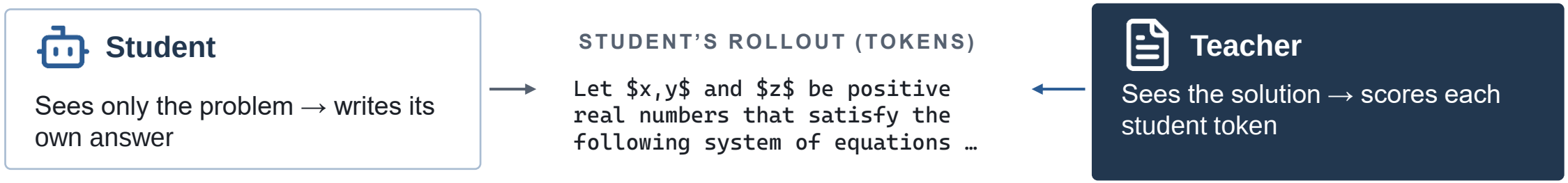
r = 0.895 Alignment–accuracy correlation



Alignment score vs. performance (merged models)

ON-POLICY SELF-DISTILLATION (OPSD)

No external teacher: the **same LLM, shown the solution**, grades its own attempt token by token.



Challenges of GRPO/OPSD & **Our solution**

✕ GRPO (RL)

Needs **many rollouts** → slow, expensive

✕ OPSD

Dense supervision; **can be unstable**

✓ S^3-OPSD (ours)

Key tokens only → **13× faster than GRPO**, AIME24 **+10.0**

Why it matters cheaper post-training lets a team specialize its own LLM more often, on its own data.

S^3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation

ICLR 2027 (submitted) | Co-first author

REFINING

Better LLM reasoning without an external teacher — stable and fast.

PROBLEM

Self-distillation is **costly & unstable**; RL needs **many rollouts**

IDEA

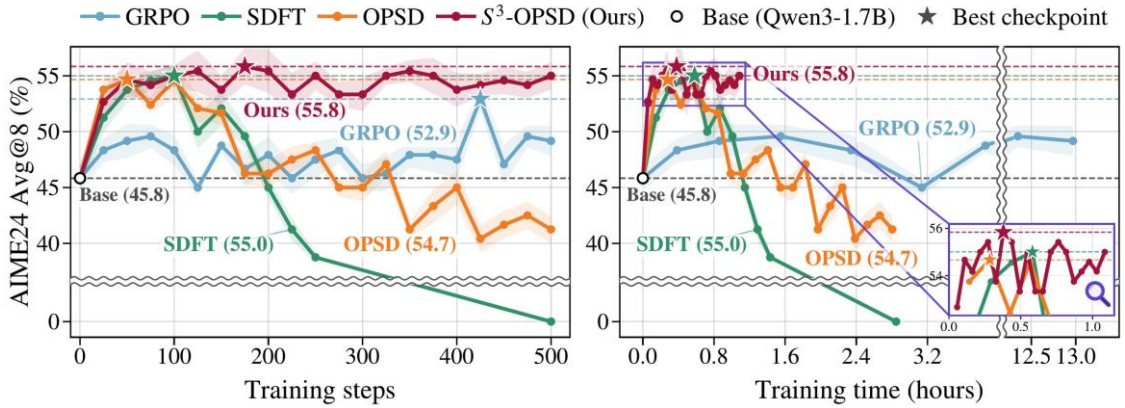
Supervise only the **important tokens**, with **small, stable updates**

RESULT

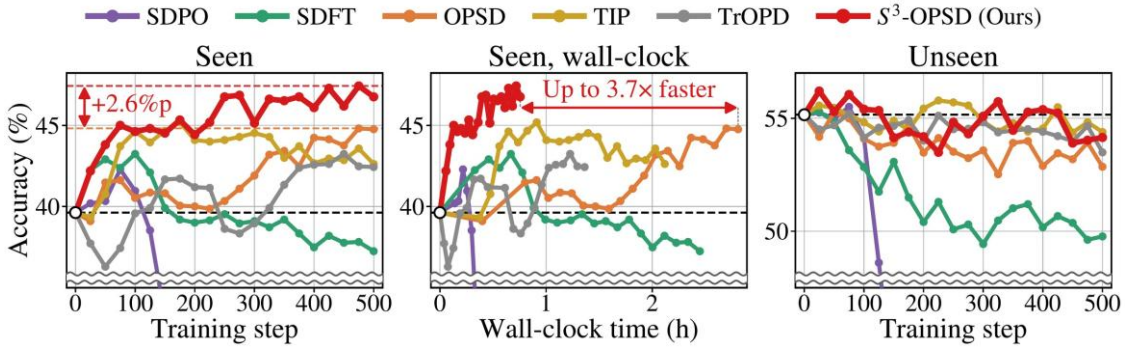
AIME24 **55.8 (+10.0)** · up to **13× faster** than RL (GRPO)

APPLICABLE TO LLM / AGENTS

Cheaper, stable **post-training of in-house LLMs**



AIME24 vs. training steps (left) and wall-clock time (right)



Multi-task learning over seven domains

Self-Distilled Reinforcement Learning for Language-Robust VLA Models

ICLR 2027 (submitted) | Co-first author

REFINING

A robot that succeeds with one instruction can fail with a paraphrase of it.

PROBLEM

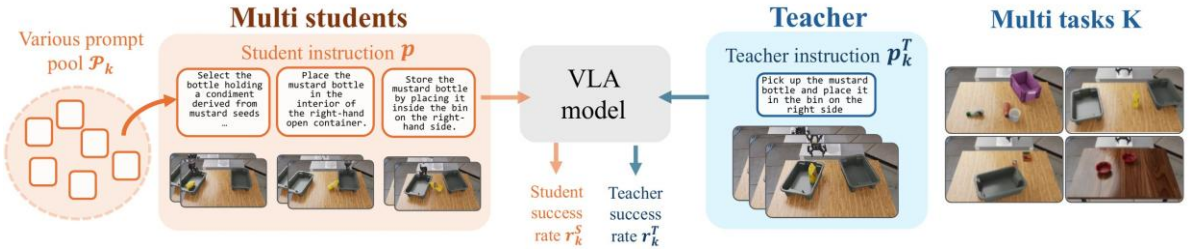
Paraphrased instructions → **very different success rates**

IDEA

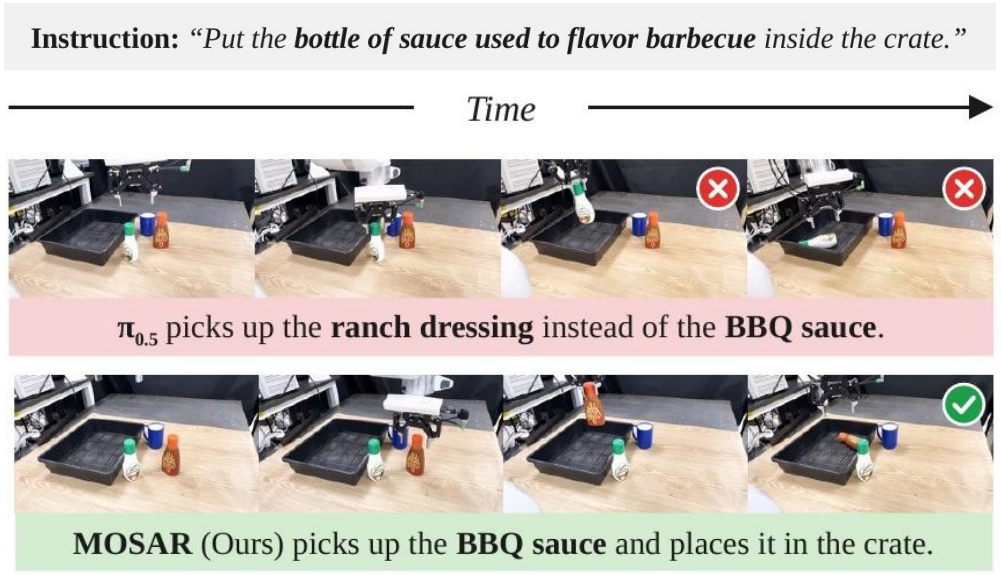
OPSD from the **best-phrased prompt** to its paraphrases; **RL** keeps improving the best

RESULT

Robust to paraphrases; unseen prompts & tasks preserved · verified on a **real Franka robot**



Overview of MOSAR



Real robot: only MOSAR picks the correct object

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CONCEPT ERASURE

REMOVE ONE THING, KEEP EVERYTHING ELSE

Generative models can reproduce **copyrighted, private or unsafe** content.

Goal: delete **only the target** — no full retraining, no damage to the rest.

STEP 1 · PRECISE

CPE

ICLR 2025 · Co-first author

A **gate** fires only for target concepts

✓ **50 celebrities / 100 styles** erased

STEP 2 · LOCALIZED

GLoCE

CVPR 2025 · Co-first author

Erase only the **target's region** — no training

✓ **Other concepts** in the image intact

STEP 3 · SCALABLE

ETC

CVPR 2026 · Co-first · Champion Award

One **mixture-of-experts** module for all targets

✓ **2,072 concepts** at once

NAIVE vs. SELECTIVE UPDATE

✗ **Naive fine-tuning:** erases the target but **damages related concepts** and image quality

✓ **Selective update:** target gone, **everything else preserved** — measured on both sides

Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate

ICLR 2025 | Co-first author

ERASING

Erases target concepts while keeping the others intact.

✕ **PROBLEM**

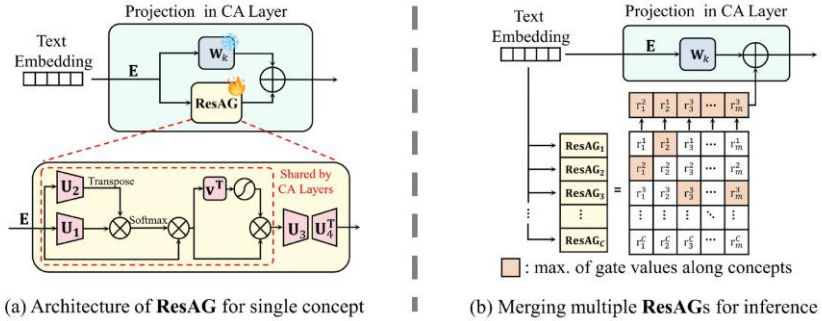
Linear fine-tuning **cannot erase and preserve at the same time**

IDEA

Add a **residual attention gate** that fires only for the targets

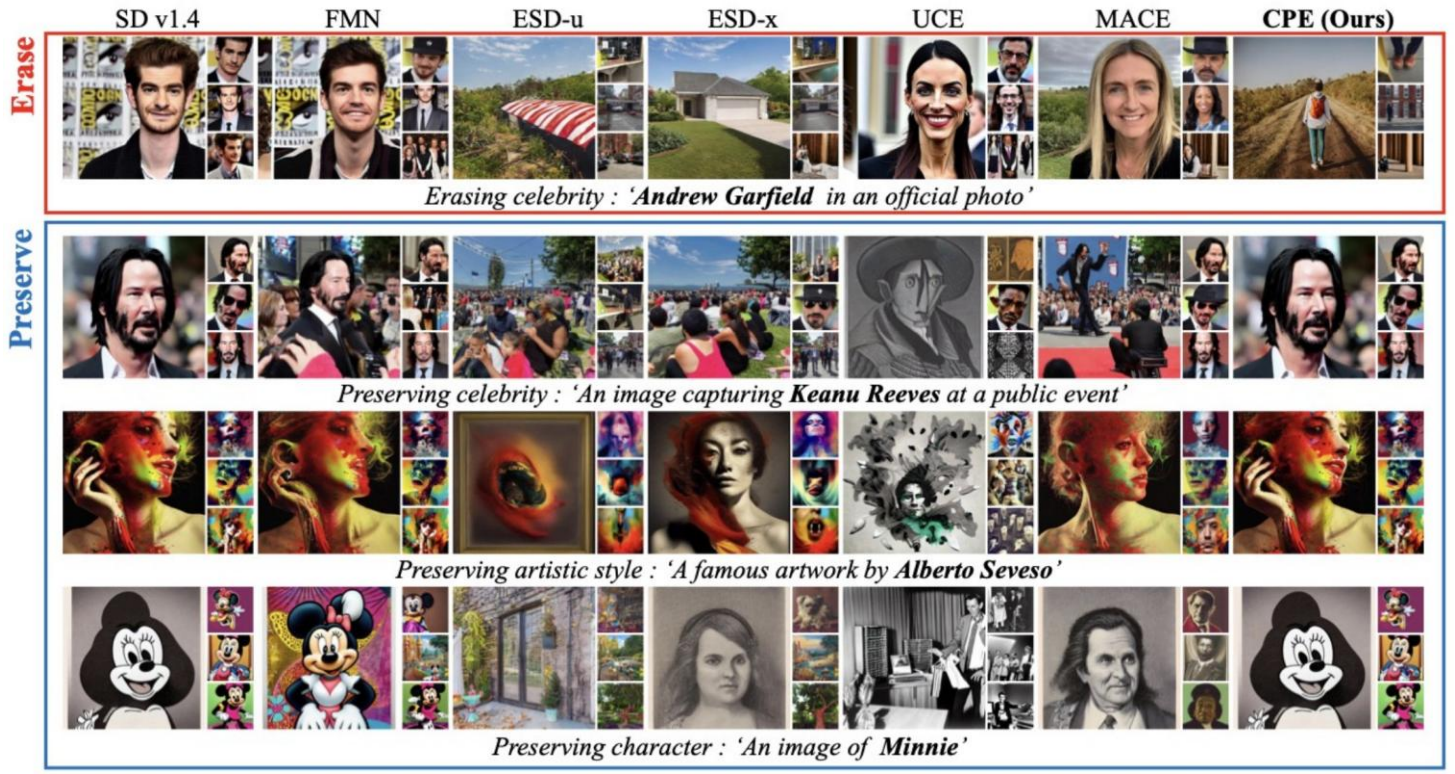
✓ **RESULT**

Erase **50 celebrities / 100 styles**, others near-original



(a) Architecture of **ResAG** for single concept

(b) Merging multiple **ResAG**s for inference



Erasing 50 celebrities: erased vs. preserved

Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation

CVPR 2025 | Co-first author

ERASING

Erases only the target's region of an image — training-free.

✕ **PROBLEM**

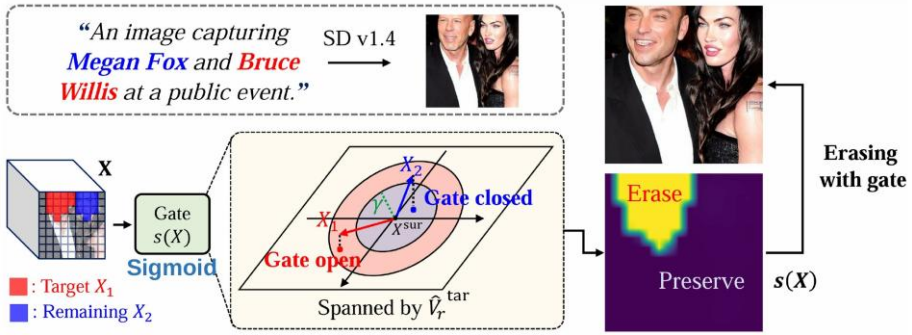
Prior methods **damage the whole image** to remove one concept

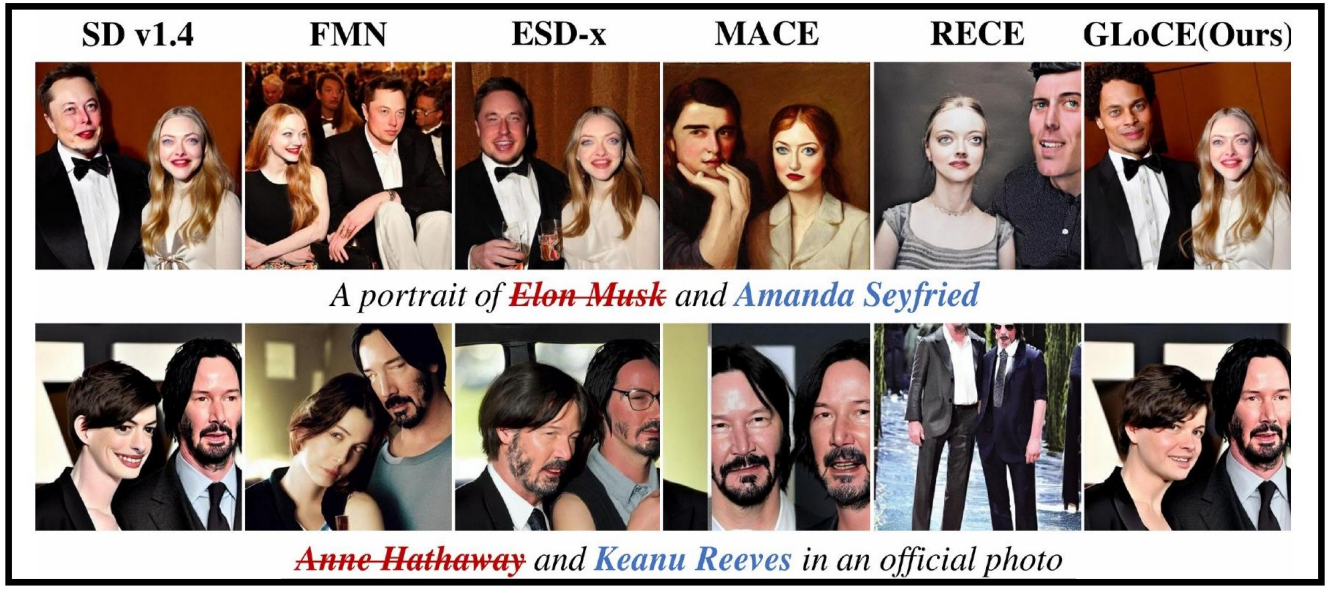
IDEA

A gate **opens only on the target region**; the rest passes untouched

✓ **RESULT**

Localized erasure on **SD v1.4 / SD v3**, **no training**





Localized celebrity erasure

Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models

CVPR 2026 | Co-first author · Compute Transparency Champion · Popular Science Korea

ERASING

Erases 2,072 concepts at once while keeping generation quality.

PROBLEM

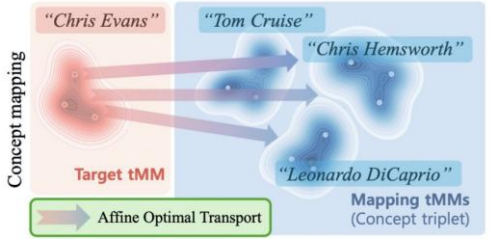
Prior methods **break down beyond a few hundred** concepts

IDEA

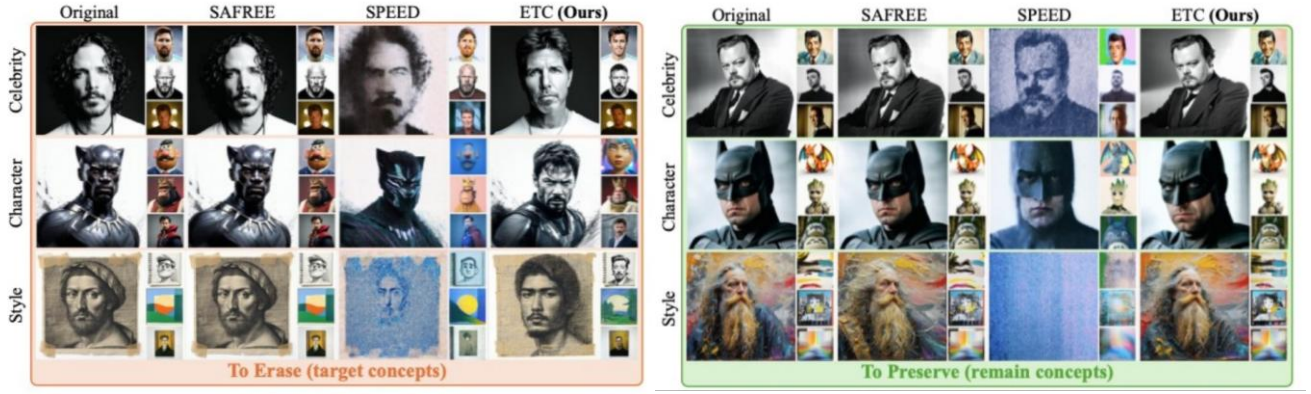
Map each target to a **safe nearby concept**; **one module** handles all

RESULT

2,072 concepts (SD v1.4) · **515** (SD v3.5)



Concept mapping via affine optimal transport



Targets erased (left) · other concepts preserved (right)

MORE ON GENERATIVE MODELS

Unlearning · Semantic Binding · Safe Diffusion

Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models

CVPR Workshop 2026 · Co-author

- Removes outputs **prompts cannot specify** (e.g., a person's face)
- **Image-editing surrogates** + timestep-aware weighting
- Validated on **SD3** and **DDPM-CelebA**

Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation

arXiv 2025 · Co-author

- Attributes often **bind to the wrong object**
- Analyzes the **geometry of text-token embeddings**
- Uses it for **stronger semantic binding**

Selective Concept Erasing for Safe Diffusion Models

KSPC 2024 · First author · Best Poster

- Erases **unsafe concepts**, keeps the rest
- **Best Poster Award**
- Early work behind **CPE, GLoCE and ETC**

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IMAGE RESTORATION & EFFICIENT VISION

Low-level vision · On-device AI

CVPR 2023 · Co-author

All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters

- One model restores **unknown degradation types**
- Granted **US (2026) and KR (2025) patents**

→ **Relevant to:** *denoising / deblurring inspection images*

ECCV Workshop 2022 · Challenge participant

Mobile AI & AIM 2022 Challenges — Quantized Super-Resolution on Mobile NPUs · Depth Estimation on Mobile

- Quantized **super-resolution on mobile NPUs**
- Co-author of the **challenge reports**

→ **Relevant to:** *fast inference on edge devices*

ICASSP 2021 · First author

Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems

- Speeds up **inverse-problem solvers** with a network

→ **Relevant to:** *physics-based image reconstruction*

KSPC 2022 · Co-author

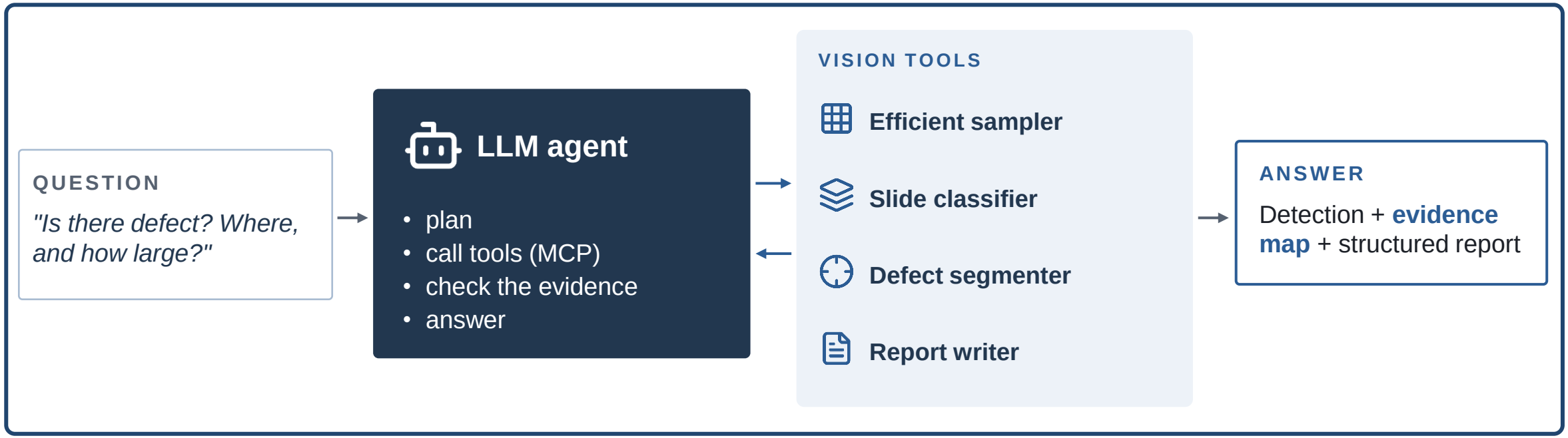
Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction

- **Uncertainty-aware** low-dose CT reconstruction

→ **Relevant to:** *reconstruction from noisy measurements*

FUTURE DIRECTION SPECIALIZED AGENTS WITH REINFORCEMENT LEARNING

RL for an agent that answers an expert's question by **calling diverse vision models as tools** with **minimal forgetting**



CONCLUSION

01 Inspection-ready vision for huge, domain-specific images

Few-shot, **training-free defect segmentation** (Samsung DS), **agentic defect analysis**, and **gigapixel image analysis** — detection, localization and reports

02 Update Models without forgetting

Selective knowledge updates — expanding, erasing and refining knowledge via **post-training** and **training-free closed-form updates**, validated at scale (2,000+ concepts, MLLMs, VLA on a real robot)

03 LLM post-training, leadership & industry collaboration

NAVER Cloud HyperCLOVA X post-training; **project leader** for SD Biosensor, MSIT · NIPA and KETI; **team leader** for MICCAI 2025 REG (2nd); projects with **Samsung DS and Samsung Research**

Next Bringing these to industrial AI: specialized agents that **absorb new processes and knowledge** using diverse tools.

Thank you

Byung Hyun Lee · AI Researcher

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